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Epistemic Bisimulation Geometry: Finite-State Certificates for Representation and Exploration

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ABSTRACT A bisimulation metric is a natural geometry for exploration only once it is known, yet learning it requires the very exploration it is meant to guide. We resolve this circularity by treating the metric as statistically uncertain. For finite discounted Markov decision processes, optimistic and pessimistic transport operators yield an anytime confidence interval that contains the true metric, and a diameter refinement sharpens it on the same event. The interval width satisfies a Bellman inequality over coupled state pairs, whose bounded majorant is a computable certificate. The upper endpoint supplies certified covers and anchor representations with explicit distortion. Under reset reachability, an algorithm that learns its own model and navigation policies delivers these outputs within an explicit interaction bound. Two lower bounds mark what geometry cannot buy: a metric with a single equivalence class may still require samples linear in the number of state-action pairs, and certifying a cover can cost two more powers of the accuracy than finding one. In eleven pre-specified studies, including RiverSwim, the certificates contained the true metric in every run, and the refinement narrowed them up to sevenfold. They certified state-aggregation decisions without error, while bootstrap intervals from the same samples wrongly separated bisimilar states in up to 87% of runs. Learned navigation cut interaction 34-fold on RiverSwim, and influence priorities shortened a targeted query by 24% where irrelevant rows were cheap. A neural instantiation learned a partly invariant encoder on a rich-observation block MDP, but not yet informative uncertainty. Code and data: https://github.com/Adityasikarwar3107/epistemic-bisimulation-geometry.

INDEX TERMS Bisimulation metrics, reinforcement learning, epistemic uncertainty, optimal transport, representation learning, exploration, sample complexity.

I. INTRODUCTION

An exploring agent has to decide which differences between observations matter. A new background texture may be irrelevant to control, while a barely visible key may change which trajectories are possible at all. Bisimulation answers this question behaviorally. Two states are close when their rewards are close and their action-conditioned successor distributions are close, where the successors are compared recursively in the same geometry [1], [2]. Representation learning uses this criterion to discard nuisance variation [3]–[5], and an explorer can use the same distances to tell behaviorally new regions apart from familiar ones.

The difficulty is statistical, and it cuts both ways. An inaccurate encoder can merge a new region with an old one, after which the agent never collects the data that would expose the merge. An explorer guided by a poorly learned geometry can just as easily spend its budget on nuisance variation. Our earlier work [6] relates bisimulation distortion to representation and coverage guarantees, but it assumes that adequate data and a navigation routine are already available. In this paper we ask what can be certified while the metric, the data, and the navigation policies are all still being learned.

The question matters outside benchmarks as well. Physical control problems such as microgrid energy management under renewable uncertainty are now routinely cast as Markov decision processes and solved by policy optimization [7], and in such settings every interaction spent on an unverified geometry has a real cost.

Our central object is an interval rather than a point estimate:

$$d_t^-(s,s') \le d_\sim(s,s') \le d_t^+(s,s'). \tag{1}$$

Its midpoint is the current estimate of the behavioral geometry, and its width $\Delta_t = d_t^+ - d_t^-$ says how much that estimate can still be trusted. A pair of states can be uncertain for two reasons. Either the pair itself has been observed too rarely, or the states that its coupled futures reach have been. A Bellman inequality on $\mathcal{S} \times \mathcal{S}$ accounts for both, and it is this recursion for the *width*, together with a computable envelope around the distance, that we regard as the main technical novelty. Section I-A places both in the context of pair-space formulations of bisimulation and of uncertainty Bellman equations.

A second question concerns what a geometric complexity bound can actually deliver. A covering number describes a metric space whose distances are already known; it is silent on which observations may be pooled before the geometry has been identified. We show that a true metric with a single equivalence class can still require $\Omega(SA\varepsilon^{-2})$ samples for a uniform certificate, so a small covering number cannot in general stand in for the number of statistically distinct state-action rows. The proof also makes clear what additional structure an intrinsic-dimension guarantee would have to exploit.

Our contributions are as follows.

1) Confidence operators on bounded symmetric costs. We prove that they contract, that their fixed points bracket the true metric, and that the upper endpoint is a pseudometric while the lower one in general is not. A diameter refinement sharpens both endpoints on the same confidence event, with or without a known support graph.
2) A pair-space majorant for the interval width, together with a discounted occupancy interpretation, a local bound in terms of escape probability, and residual certificates for numerical solutions. The adjoint of the majorant yields influence priorities for sampling, either for the full certificate or for a single targeted query.
3) Certified covers, a constructive anchor representation, and an algorithm that learns its model, resolves the metric, and learns its navigation policies from interaction alone, under explicit reset and reachability conditions.
4) A lower bound showing that the geometry of the true metric alone cannot control the sample cost, and a matching separation between coverage and resolution costs on a grid family indexed by resolution.
5) Exact invariance of the metric to exogenous state, and conditions under which uncertainty vanishes on resolved stochastic regions.
6) Eleven pre-specified studies. They measure interval coverage and the sharpness of the majorant, compare the certificate with a simulation-lemma interval and with bootstrap intervals computed from the same samples, test coverage, width, and compression on models with up to 128 labels and on the RiverSwim benchmark, quantify the refinement and the merge and split decisions it certifies, and separate the effects of navigation, planner tie-breaking, and priorities on acquisition.
7) A neural instantiation of the same targets (Appendix F), evaluated on a rich-observation block MDP with a strong exogenous distractor, together with a diagnostic of the small-sample bias that all sampled bisimulation targets share.

The code, the experiment specifications, and the seed-level records of every study are publicly available [8].

We should be clear about what the finite-state guarantee does and does not assume. It needs neither a trained encoder, nor replay coverage, nor a supplied navigation policy. It does need known state labels and the existence of reaching policies, and we state both explicitly. Rich observations call for a separate decoding argument, and a Euclidean encoder carries its own approximation term. We keep these dependencies in view throughout, because a learning loop can only be closed when each of them is either earned by the algorithm or assumed in the open.

A. RELATED WORK

Computing and estimating bisimulation metrics. Ferns et al. [9] already approximated the Kantorovich terms of the metric by statistical sampling, and Castro [10] gave a sampling scheme that converges to a policy-dependent metric in deterministic MDPs. More recently, Calo et al. [11] estimate distances between finite Markov chains from sample streams, with guarantees on both computation and sample complexity. Pair-space formulations are not new either: Ferns and Precup [12] identify bisimulation distances with the values of coupled control problems. All of these works produce a point estimate of a distance, accompanied by a bound on its error. What we return instead is a simultaneous, anytime interval for a finite MDP, a pair-space majorant that explains where its width comes from, and a certified cover reached by learned navigation. Our contribution is therefore not the first statistical estimator of bisimulation distances; it is the interval, the recursion for its width, and the closed loop built on top of them.

Interval models and optimism. Bounded-parameter MDPs [13] propagate interval bounds on rewards and transitions through optimistic and pessimistic value iteration. Model-based interval estimation (MBIE) [14] and UCRL2 [15] build confidence sets on rows to drive optimistic exploration. Our operators $\mathcal{T}^{\pm}$ play the role of interval value iteration for a behavioral distance, with two differences that matter. The ground cost of the transport term is itself the unknown fixed point, and the lower iterate is in general not a metric, so stability has to be proved without the triangle inequality. Uncertainty Bellman

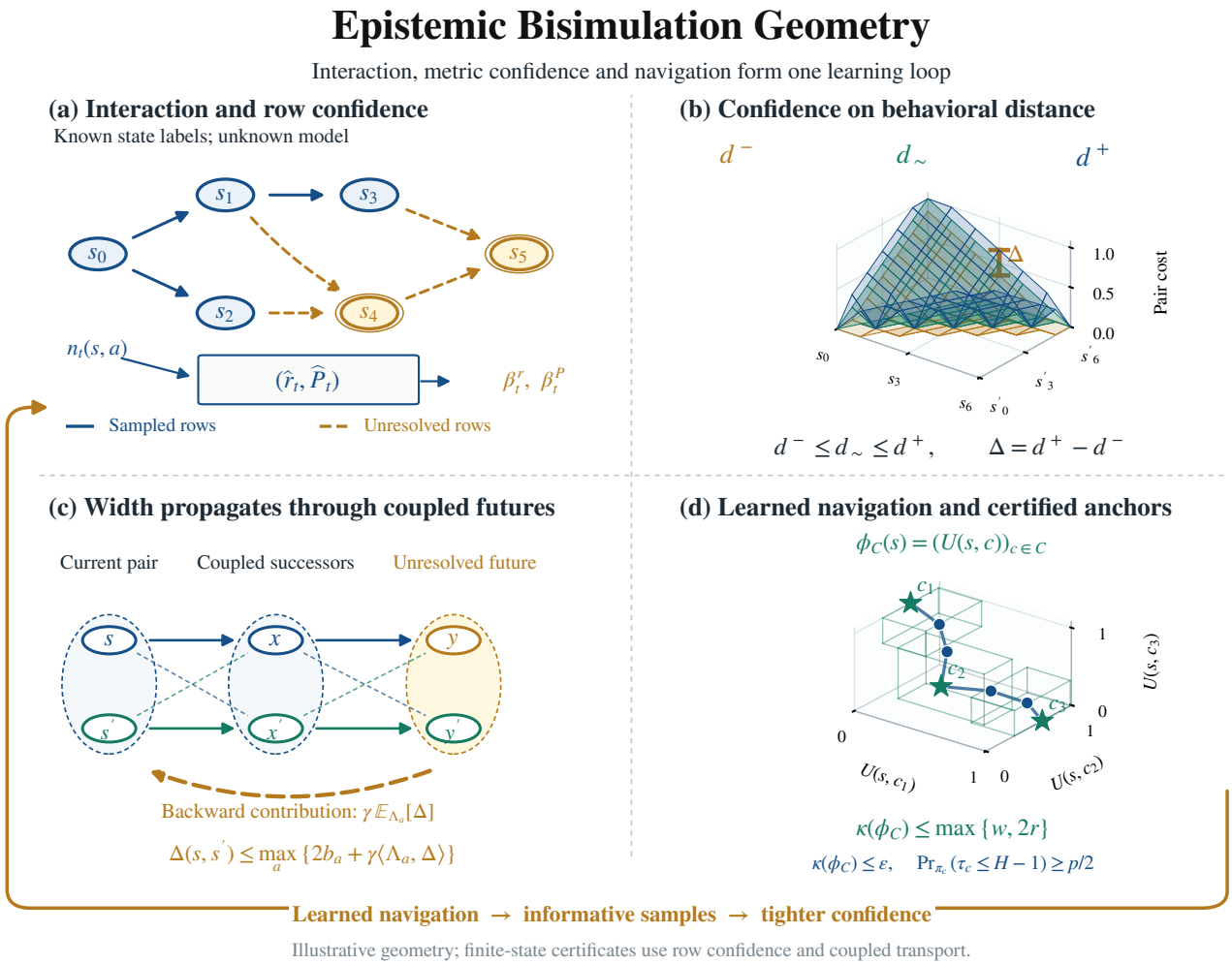


FIGURE 1: Epistemic bisimulation geometry and its learning loop. (a) Interaction supplies row counts, empirical models, and confidence radii; no accurate encoder is assumed. (b) A bounded interval of pair costs encloses the behavioral distance. (c) Forward transport couplings carry uncertainty backwards from unresolved successor pairs. (d) The anchor map is built from the final upper endpoint U; wire boxes show ℓ_∞ covering neighborhoods. The gold path feeds learned navigation back into informative interaction. Surfaces and coordinates are illustrative constructions, not measurements, and do not assume exact Euclidean realizability.

equations propagate posterior uncertainty about values [16]; in our setting the propagated quantity is the width of a distance interval, and the transport coupling is part of the construction.

Exploration with a learning guarantee. The planner over known and unknown rows follows the optimism principle of E^3 and R-MAX [17], [18]. Reward-free exploration [19], [20] likewise learns a model that is uniformly accurate, so that any reward supplied later can be planned for. Our closed-loop theorem shares this reward-free character, but its target is metric accuracy and a certified representation rather than near-optimal planning, and its navigation policies are learned rather than given. Propagated priorities change how samples are allocated, yet the worst-case guarantee holds for any priorities in $[1/2, 1]$, which is why Section VIII measures their effect separately.

Bisimulation for representation and exploration. Deep bisimulation methods learn encoders whose distances approximate a behavioral metric [3]–[5], [21], and robustness analyses show how such objectives can collapse or explode when the dynamics model is underconstrained [22]. LIBERTY builds potential-based exploration rewards from an inverse-dynamics bisimulation metric [23], and TEB counters metric collapse under sparse rewards with predicted reward differences [24]. These methods modify the metric or the reward, and their guarantees concern shaping or representation quality. We study confidence intervals for the fixed, max-over-actions metric. Autonomous exploration methods learn goal-reaching policies without a predefined navigation routine [25], [26], and their reachability analysis motivates our stopped, optimistic exploration model. Latent-state decoding addresses perceptual

identification [27]–[30], a problem complementary to ours, and RichCLD shows that smoothness assumptions which look alike can have quite different statistical consequences under rich observations [31]. For this reason we never infer perceptual identification from a small bisimulation covering number.

II. SETTING AND MATHEMATICAL PRELIMINARIES

A. DISCOUNTED BEHAVIORAL GEOMETRY

Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ be a stationary finite MDP with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, $0 \le \gamma < 1$, a declared bound $R_{\max} > 0$, and mean rewards $r(s,a) \in [0, R_{\max}]$. Reward observations lie in the same interval, and we write $P_{s,a} = P(\cdot \mid s, a)$.

For a bounded nonnegative cost h, the primal transport problem is

$$\mathrm{OT}_h(\mu,\nu) = \min_{\Lambda \in \Pi(\mu,\nu)} \langle \Lambda, h \rangle, \tag{2}$$

where $\Pi(\mu,\nu)$ is the set of nonnegative matrices with row marginal μ and column marginal ν, and $\langle \Lambda, h \rangle = \sum_{x,y} \Lambda(x,y) h(x,y)$. When h is a pseudometric, $\mathrm{OT}_h(\mu,\nu) = W_1(\mu,\nu;h)$. We nevertheless work with the primal form throughout, because several of the costs produced by confidence bounds violate the triangle inequality.

The normalized bisimulation operator is

$$\mathcal{T}[h](s,s') = \max_{a \in \mathcal{A}} \{ (1-\gamma)|r(s,a) - r(s',a)| + \gamma\, \mathrm{OT}_h(P_{s,a}, P_{s',a}) \}. \tag{3}$$

Its unique fixed point is the bisimulation pseudometric $d_\sim$, which satisfies $0 \le d_\sim \le R_{\max}$ [1], [2]. Distinct state labels may lie at distance zero, and the quotient of the pseudometric identifies them.

Definition 1 (Representation distortion)**.** For a representation ϕ with values in a normed space,

$$\kappa(\phi; X) = \sup_{s,s' \in X} \big| \, \|\phi(s) - \phi(s')\| - d_\sim(s,s') \big|. \tag{4}$$

For a specified Euclidean encoder class Φ, let $\kappa_\Phi = \inf_{\phi \in \Phi} \kappa(\phi; \mathcal{S})$.

Every theorem that involves a representation states its norm, and the choice is not cosmetic. A finite pseudometric always admits an exact ℓ_∞ anchor embedding, but it need not admit an exact low-dimensional Euclidean one, so κ_Φ cannot in general be assumed to vanish.

B. COVERING, DIMENSION, AND CONTROL

Let $\mathcal{N}_\sim(\varepsilon; X)$ denote the smallest cardinality of an ε-cover of X in $d_\sim$. The doubling constant is the smallest λ such that every ball in the quotient is covered by at most λ balls of half the radius, and the doubling dimension is $D = \log_2 \lambda$. On a bounded doubling space,

$$\mathcal{N}_\sim(\varepsilon; X) \le C \left(\frac{\mathrm{diam}_\sim(X)}{\varepsilon} \right)^D \tag{5}$$

over the nontrivial range of scales, with a constant C that depends on the geometry. A slope fitted to covering numbers on a log–log plot estimates this exponent over a finite range of scales, but it does not define the doubling dimension.

Reachability is a property of transitions and target events, and reward equivalence alone does not determine it. We analyze episodes that reset to s_0. The finite-state theorem assumes a known label set and the existence of policies that reach each label with a stated probability within a stated horizon; the algorithm has to learn these policies itself. Resets are excluded from the interaction count when they are free and counted explicitly when they have a physical cost.

C. ANYTIME MODEL CONFIDENCE

After $n(s,a)$ retained observations of a row, let $\hat{r}(s,a)$ and $\hat{P}_{s,a}$ denote the empirical estimates. Let Γ be a known upper bound on the number of possible successors of any state-action row; $\Gamma = S$ is always valid. For $n \ge 1$ we set

$$\ell_n = \log \frac{4SA\pi^2 n^2}{3\delta}, \tag{6}$$

$$\beta_n^r = R_{\max} \min \left\{ 1, \sqrt{\frac{\ell_n}{2n}} \right\},$$

$$\beta_n^P = \min \left\{ 1, \sqrt{\frac{\Gamma \log 2 + \ell_n}{2n}} \right\}. \tag{7}$$

With no observations we take $\beta_0^r = R_{\max}$, $\beta_0^P = 1$, and any bounded empirical reward and probability vector. Total variation is $\mathrm{TV}(p,q) = \frac{1}{2}\|p - q\|_1$.

Lemma 2 (Simultaneous confidence)**.** *With probability at least $1-\delta$, at every visit count and for every state-action pair,*

$$|r - \hat{r}| \le \beta^r, \qquad \mathrm{TV}(P, \hat{P}) \le \beta^P. \tag{8}$$

The lemma follows from Hoeffding's inequality and the multinomial ℓ_1 deviation inequality [32], [33], combined with a union bound over rows and visit counts (Appendix A). Because the confidence is indexed by the number of visits to each row, it remains valid under adaptive interaction. One detail deserves emphasis: Γ bounds the *possible* successors, including those not yet observed, and replacing it by the number of observed successors is invalid unless a correction for the missing mass is added.

D. THE CORRECT VALUE GUARANTEE

Proposition 3 (Optimal values and policy metrics)**.** *For the unnormalized discounted optimal value,*

$$|V^*(s) - V^*(s')| \le \frac{d_\sim(s,s')}{1-\gamma}. \tag{9}$$

For a fixed state-dependent policy π, let $r^\pi(s) = \sum_a \pi(a \mid s) r(s,a)$ and $P_s^\pi = \sum_a \pi(a \mid s) P_{s,a}$. The single-action bisimulation metric d^π of this induced model satisfies $|V^\pi(s) - V^\pi(s')| \le d^\pi(s,s')/(1-\gamma)$.

The first bound follows by iterating the normalized optimal Bellman operator on functions that are 1-Lipschitz in $d_\sim$. It

TABLE 1: Principal symbols. Statistical state complexity and geometric covering complexity are kept distinct.

Symbol	Meaning
S, A, Γ	Labels, actions, successor support bound
$d_\sim$	True max-over-actions bisimulation metric
d^-, d^+	Pessimistic cost and optimistic metric
$\Delta, \bar{d}$	Interval width and midpoint
b_a	Local pairwise uncertainty source
Λ^a, U_c	Lower-cost coupling and bounded pair majorant
$\langle \Lambda, h \rangle$	Coupling expectation $\sum_{x,y} \Lambda(x,y)h(x,y)$
$\Sigma_{s,a}, D_{s,a}$	Successor labels of a row and their diameter bound
ρ, ι_w	Pair occupancy and row influence (adjoint)
L, U	Returned certified lower and upper endpoints
H, p	Reset horizon and reachability probability
n_0	Retained samples required per row
$D, \mathcal{N}_\sim$	Doubling dimension and covering number
κ_Φ	Euclidean representation approximation floor
η_K	Empirical probability of escaping K

does not extend to V^π for an arbitrary policy, since such a policy may act differently at bisimilar labels. The distinction is easy to overlook in continuous-action experiments, where a policy metric is often used in place of the max-over-actions metric.

III. CONFIDENCE OPERATORS FOR BEHAVIORAL DISTANCES

A. A SPACE OF COSTS, NOT ONLY METRICS

We work on the complete space $\mathfrak{C}$ of costs $h : \mathcal{S}^2 \to [0, R_{\max}]$ that are symmetric, $h(s,s') = h(s',s)$, and vanish on the diagonal, $h(s,s) = 0$, equipped with the supremum norm. At a fixed data snapshot, $\beta^r(s,a)$ and $\beta^P(s,a)$ denote the row confidence radii. For $s \neq s'$, define

$$\begin{aligned} e^r_a(s,s') &= \beta^r(s,a) + \beta^r(s',a), \\ e^P_a(s,s') &= \beta^P(s,a) + \beta^P(s',a), \\ b_a(s,s') &= (1-\gamma)e^r_a(s,s') + \gamma R_{\max} e^P_a(s,s'), \end{aligned} \tag{10}$$

and set $b_a(s,s) = 0$. For a fixed off-diagonal pair (s,s'), which we suppress from the notation, let

$$\begin{aligned} \hat{g}_a &= |\hat{r}(s,a) - \hat{r}(s',a)|, \\ \hat{w}_a(h) &= \mathrm{OT}_h(\hat{P}_{s,a}, \hat{P}_{s',a}), \\ q_a(h) &= (1-\gamma)\hat{g}_a + \gamma \hat{w}_a(h). \end{aligned} \tag{11}$$

The optimistic and pessimistic operators are then

$$\mathcal{T}^+[h] = \min\left\{R_{\max}, \max_a [q_a(h) + b_a]\right\}, \tag{12}$$

$$\begin{aligned} \mathcal{T}^-[h] = \max_a \big\{ &(1-\gamma)\left(\hat{g}_a - e^r_a\right)_+ \\ &+ \gamma\left(\hat{w}_a(h) - R_{\max} e^P_a\right)_+ \big\}. \end{aligned} \tag{13}$$

Both operators vanish on the diagonal, as they should: uncertainty about a state's model ought not to create a distance between that state and itself.

Lemma 4 (Transport stability for arbitrary bounded costs)**.** *For $h, \tilde{h} \in [0, R_{\max}]^{S \times S}$,*

$$\left|\mathrm{OT}_h(\mu,\nu) - \mathrm{OT}_{\tilde{h}}(\mu,\nu)\right| \le \left\|h - \tilde{h}\right\|_\infty, \tag{14}$$

$$\begin{aligned} \left|\mathrm{OT}_h(\mu,\nu) - \mathrm{OT}_h(\tilde{\mu},\tilde{\nu})\right| \le{}& R_{\max}\,\mathrm{TV}(\mu,\tilde{\mu}) \\ &+ R_{\max}\,\mathrm{TV}(\nu,\tilde{\nu}). \end{aligned} \tag{15}$$

Transport is monotone in its ground cost.

The proof uses feasible couplings and a maximal coupling of the perturbed marginals; it needs neither duality nor the triangle inequality (Appendix B). This is precisely what lets us apply the same estimates to the nonmetric iterates of $\mathcal{T}^-$.

Theorem 5 (Epistemic sandwich)**.** *The operators $\mathcal{T}^+$ and $\mathcal{T}^-$ map $\mathfrak{C}$ into itself, are monotone, and are γ-contractions. On the event of Lemma 2, their unique fixed points d^+ and d^- satisfy, simultaneously at all data snapshots,*

$$d^- \le d_\sim \le d^+. \tag{16}$$

The upper endpoint d^+ is a pseudometric. The lower endpoint is in general only a bounded symmetric cost.

Proof. Monotonicity and contraction follow from Lemma 4, since positive parts, maxima, and clipping are nonexpansive. On the confidence event, the true reward difference lies between $(|\hat{r} - \hat{r}'| - e^r_a)_+$ and $|\hat{r} - \hat{r}'| + e^r_a$, and (15) gives the corresponding bounds for the transport term. Hence $\mathcal{T}^-[h] \le \mathcal{T}[h] \le \mathcal{T}^+[h]$ for every $h \in \mathfrak{C}$. Iterating the outer operators from $d_\sim$ and using monotonicity and contraction gives the sandwich. That d^+ is a pseudometric is proved in Appendix B-B. □

The asymmetry between the two endpoints is genuine. Take $\gamma = 0$, rewards $(0, \frac{1}{2}, 1)$, and reward radii $(0, \frac{1}{2}, 0)$. Then $d^-(1,2) = d^-(2,3) = 0$ while $d^-(1,3) = 1$, which violates the triangle inequality; the midpoint with the upper endpoint need not be a metric either. Both remain perfectly legitimate as interval bounds and as regression targets, but neither should be used where a metric is required.

Figure 2 makes the same point geometrically. For three labels, a symmetric cost with zero diagonal has three coordinates (h_{12}, h_{23}, h_{13}). The triangle inequalities cut a bounded metric cone out of the cost cube, whereas the confidence interval produced by the operators is a coordinate box. That the box contains the true metric does not make every cost inside it a pseudometric.

B. COMPARISON WITH MODEL-SPACE UNCERTAINTY

Let $\mathcal{V}_t$ be the set of MDPs that satisfy the row confidence constraints at a snapshot, and write $d_{\mathcal{M}'}$ for the exact metric of $\mathcal{M}' \in \mathcal{V}_t$. The pointwise model-space envelope is

$$L_{\mathcal{V}} = \inf_{\mathcal{M}' \in \mathcal{V}_t} d_{\mathcal{M}'}, \qquad U_{\mathcal{V}} = \sup_{\mathcal{M}' \in \mathcal{V}_t} d_{\mathcal{M}'}. \tag{17}$$

The operator comparison used above also shows that

$$d^- \le L_{\mathcal{V}} \le d_\sim \le U_{\mathcal{V}} \le d^+. \tag{18}$$

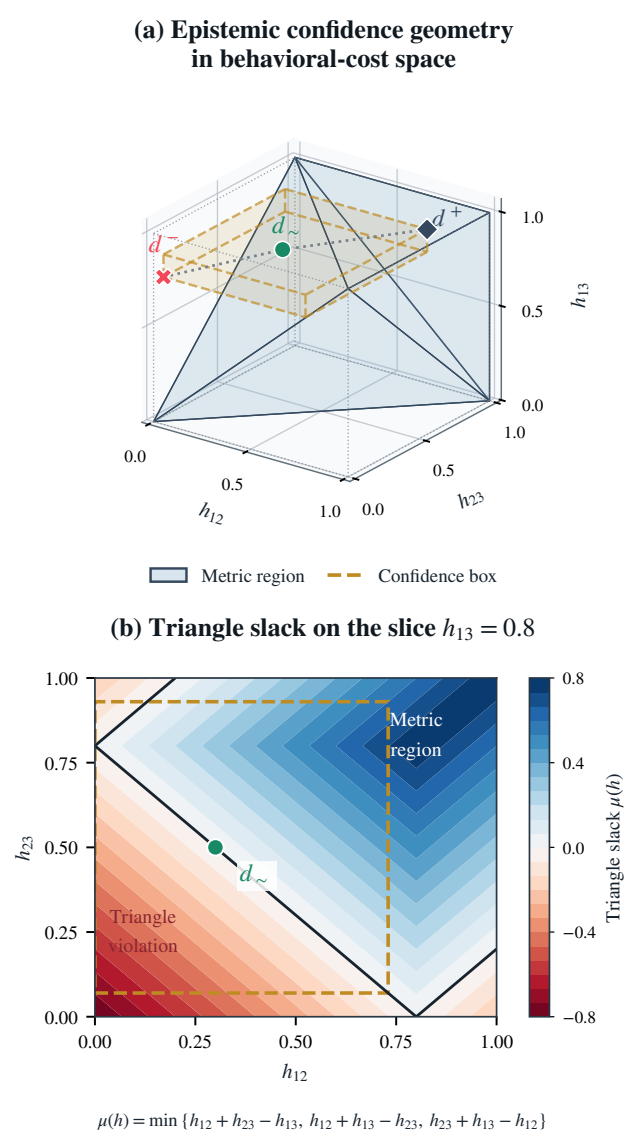


FIGURE 2: Geometry of the epistemic sandwich. (a) Three-state costs in $[0,1]^3$, their triangle-valid region, and the coordinate confidence box. With $\gamma = 0$, exact empirical rewards $(0, 0.3, 0.8)$ and reward radii $(0.03, 0.4, 0.03)$ give $d^- = (0, 0.07, 0.74)$, $d_\sim = (0.3, 0.5, 0.8)$, and $d^+ = (0.73, 0.93, 0.86)$ in pair order $(12, 23, 13)$. The lower cost violates a triangle inequality by 0.67. (b) The slice $h_{13} = 0.8$. Nonnegative triangle slack characterizes the metric region; the dashed rectangle is the confidence box restricted to this slice. The construction is analytical and separate from the experiments.

The computable operators therefore give an outer envelope, which need not be the tightest interval compatible with a single model. The source of the conservatism is easy to name. Each pair is optimized on its own, so the consistency constraints that a shared model imposes across pairs are ignored.

C. METRIC UNCERTAINTY PROPAGATES ON STATE PAIRS

For each off-diagonal pair and action, select an optimal plan for the lower cost,

$$\Lambda^a_{s,s'} \in \operatorname*{arg\,min}_{\Lambda \in \Pi(\hat{P}_{s,a}, \hat{P}_{s',a})} \langle \Lambda, d^- \rangle. \tag{19}$$

Reversed pairs use the transposed plan, and a diagonal pair uses the diagonal coupling of $\hat{P}_{s,a}$ with itself.

Theorem 6 (Pair-space uncertainty Bellman inequality)**.** *Let* $\Delta = d^+ - d^-$. *Then*

$$\Delta(s,s') \le \max_a \left\{ 2b_a(s,s') + \gamma \langle \Lambda^a_{s,s'}, \Delta \rangle \right\}. \tag{20}$$

The map on the right-hand side, denoted $\mathcal{F}$, *has a unique nonnegative bounded fixed point* U, *and*

$$\Delta \le \min\{R_{\max}, U\},$$
$$\|\Delta\|_\infty \le \min\left\{ R_{\max}, \frac{2\,\|b\|_\infty}{1-\gamma} \right\}. \tag{21}$$

Consequently, for $\bar{d} = (d^+ + d^-)/2$,

$$|\bar{d} - d_\sim| \le \tfrac{1}{2}\Delta. \tag{22}$$

Proof. Dropping the clipping gives $d^+ \le \max_a(q_a(d^+) + b_a)$, and the two positive parts in $\mathcal{T}^-$ give $d^- \ge \max_a(q_a(d^-) - b_a)$. Subtracting and using $\max f - \max g \le \max(f - g)$ reduces the difference to a transport term. Since $\Lambda^a_{s,s'}$ is feasible for the upper cost,

$$\mathrm{OT}_{d^+}(\hat{P}, \hat{P}') - \mathrm{OT}_{d^-}(\hat{P}, \hat{P}') \le \langle \Lambda^a_{s,s'}, \Delta \rangle, \tag{23}$$

which proves (20). Because $\mathcal{F}$ is monotone and contractive, $\Delta \le U$. Taking supremum norms gives (21), and the midpoint bound follows from the sandwich. □

The theorem is an inequality, and it should be read as one. Different maximizing actions, clipping, and the choice of transport plan can each make the majorant loose. In particular, a small width does not imply that every coupled future has been sampled. What the theorem provides is a sufficient certificate, not an equivalence between width and visitation.

Proposition 7 (Bounded pair majorant)**.** *Let* $\mathcal{F}_c[h] = \min\{R_{\max}, \mathcal{F}[h]\}$ *pointwise, with zero diagonal. This map is monotone and* γ*-contractive, and its fixed point* U_c *satisfies*

$$\Delta \le U_c \le \min\{R_{\max}, U\}. \tag{24}$$

Proof. Clipping is monotone and nonexpansive. Since $\Delta \le R_{\max}$ and $\Delta \le \mathcal{F}[\Delta]$, iterating $\mathcal{F}_c$ from Δ produces an increasing sequence that converges to U_c. Iterating from zero stays below both U and $R_{\max}$, which gives the remaining inequalities. The diagonal pair is absorbing with zero source. □

Clipping inside the recursion can be strictly sharper than clipping the unbounded solution after convergence. Numerical errors are handled as follows. An approximation with uniform residual ρ_W lies within $\rho_W/(1-\gamma)$ of U_c. A plan optimized

under an approximation of the lower cost with uniform error e_- is suboptimal under d^- by at most $2e_-$, and adding $2\gamma e_-$ to the off-diagonal source absorbs this error. Endpoint padding is accounted for separately whenever computed widths are compared with the majorant.

Corollary 8 (Discounted occupancy representation)**.** *For a stationary pair policy σ, let P^σ_Λ be its pair transition matrix and b^σ its local source. Then*

$$U(s,s') = \sup_\sigma \mathbb{E}^\sigma_{s,s'} \sum_{j\geq 0} 2\gamma^j b_{A_j}(X_j, Y_j), \tag{25}$$

$$U^\sigma = 2(I - \gamma P^\sigma_\Lambda)^{-1} b^\sigma. \tag{26}$$

In words, the majorant is the largest discounted amount of local uncertainty that a coupled pair can accumulate along its future. The pair process is a mathematical device, and an agent cannot impose this coupling on two independent real trajectories. Collecting the data that reduce U therefore requires a separate navigation argument, which Section V supplies.

Figure 3 illustrates the mechanism on a two-state pair process. A pair with zero local source still inherits a nonzero majorant from an unresolved coupled future. Reducing the future source can bring the majorant below a requested width, and the width is then certified. A majorant above the requested budget, on the other hand, says nothing either way about whether the actual width meets it.

D. RESOLVED REGIONS AND UNRESOLVED FUTURES

Proposition 9 (Nearly closed known set)**.** *For $K \subseteq \mathcal{S}$, define*

$$b_K = \max_{s,s'\in K, a} b_a(s,s'), \qquad \eta_K = \max_{s\in K, a} \hat{P}_{s,a}(\mathcal{S}\setminus K). \tag{27}$$

Then

$$\sup_{K\times K} \Delta \leq \min\left\{ R_{\max}, \frac{2b_K + 2\gamma R_{\max}\eta_K}{1-\gamma} \right\}. \tag{28}$$

The term $2\eta_K$ may be replaced by the smaller quantity $\max_{s,s'\in K,a} \Lambda^a_{s,s'}((K\times K)^c)$.

Proof. A feasible pair coupling leaves $K\times K$ with probability at most the sum of the two marginal escape probabilities, hence at most $2\eta_K$. In (20), bound the width inside $K\times K$ by its maximum and outside by $R_{\max}$, and rearrange. □

Local sample counts alone cannot remove the uncertainty of a region whose future escapes to unresolved states. The escape term quantifies this dependence, and it is the reason we make no claim of exhausted novelty without a closure condition.

E. NUMERICAL AND SEQUENTIAL CERTIFICATES

Proposition 10 (Residual and solver error)**.** *If $\tilde{d}^\pm$ have exact-operator residuals at most $\rho_\pm$, then*

$$\left\| \tilde{d}^\pm - d^\pm \right\|_\infty \leq \frac{\rho_\pm}{1-\gamma}. \tag{29}$$

If each computed transport value carries a certified error of at most ν, replace $\rho_\pm$ by $\rho_\pm + \gamma\nu$. Subtracting this margin from the lower solution, adding it to the upper solution, and clipping to $[0, R_{\max}]$ off the diagonal preserves the statistical sandwich.

The result follows from

$$\left\| \tilde{d} - d^\pm \right\|_\infty \leq \left\| \tilde{d} - \mathcal{T}^\pm \tilde{d} \right\|_\infty + \gamma \left\| \tilde{d} - d^\pm \right\|_\infty.$$

Since only the contraction of the exact operator is used, the bound applies to any solver, including an approximate neural one.

Because the empirical model changes with the data, widths need not shrink at every snapshot. A monotone summary is nonetheless available. For valid historical endpoints, set $L_t = \max_{j\leq t} d^-_j$ and $B_t = \min_{j\leq t} d^+_j$ pointwise. A minimum of pseudometrics need not be a pseudometric, so we take its path closure

$$U_t(s,s') = \min_{s=x_0,\ldots,x_m=s'} \sum_{i=1}^m B_t(x_{i-1}, x_i). \tag{30}$$

Proposition 11 (Monotone geometric envelope)**.** *At all times $L_t \leq d_\sim \leq U_t \leq B_t$. The function U_t is a pseudometric and is nonincreasing in t, L_t is nondecreasing, and their difference is a monotone confidence width.*

By the triangle inequality, every path sum bounds $d_\sim(s,s')$ from above, and taking the minimum proves validity; the remaining claims are standard properties of shortest-path closure. The summary is computed by an all-pairs shortest-path algorithm, and it provides a navigation distance without ever treating a nonmetric midpoint as one.

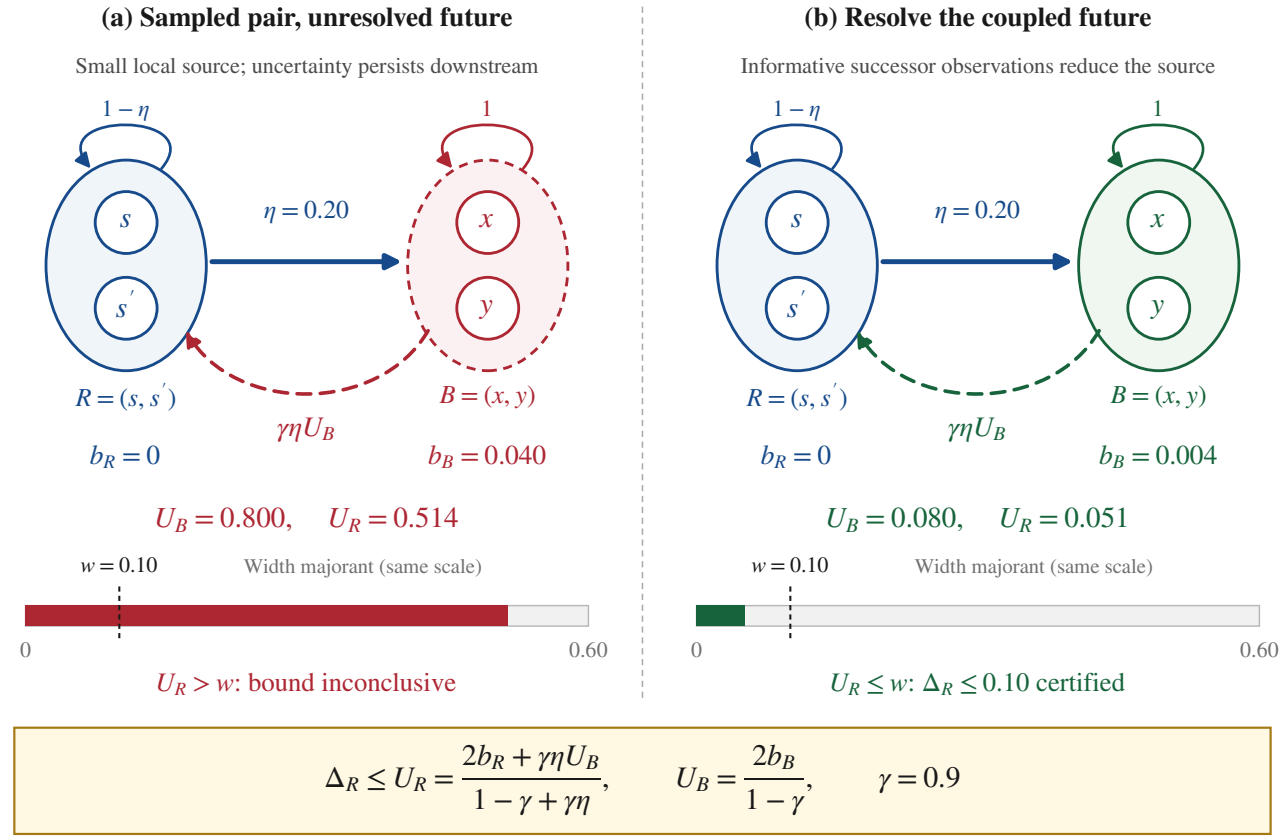
F. SHARPER ENDPOINTS ON THE SAME EVENT

Two parts of the construction are looser than they need to be, and both can be tightened without spending any additional failure probability. The first is the reward interval: any interval that is valid on the event of Lemma 2 may replace the Hoeffding interval. For observations in $[0, R_{\max}]$, the Chernoff–Hoeffding bound [32] gives the interval $\{q : n\,\mathrm{kl}(\hat{r}/R_{\max}, q/R_{\max}) \leq \ell_n\}$, whose two tails carry the same budget $2e^{-\ell_n}$ and which lies inside the Hoeffding interval by Pinsker's inequality. The second is the factor $R_{\max}$ in (15), which is merely the range of the cost. At a pseudometric cost it can be replaced by the diameter of the successor labels, and the true metric is, of course, a pseudometric.

For a row (s,a), let $\Sigma_{s,a} \subseteq \mathcal{S}$ contain the supports of $P_{s,a}$ and $\hat{P}_{s,a}$. Given reward intervals $[\underline{r}(s,a), \bar{r}(s,a)]$, write $\bar{r}_s = \bar{r}(s,a)$ and $\underline{r}_s = \underline{r}(s,a)$ for a fixed action, and define, for constants $D_{s,a}$, the gap bounds and the transport term

$$\begin{aligned} \bar{g}_a(s,s') &= \max\{\bar{r}_s - \underline{r}_{s'}, \bar{r}_{s'} - \underline{r}_s\}, \\ \underline{g}_a(s,s') &= (\max\{\underline{r}_s - \bar{r}_{s'}, \underline{r}_{s'} - \bar{r}_s\})_+, \\ \pi_a(s,s') &= \gamma\{D_{s,a}\beta^P(s,a) + D_{s',a}\beta^P(s',a)\}. \end{aligned} \tag{31}$$

Proposition 12 (Refined endpoints)**.** *Suppose that on an event $\mathcal{E}$ the reward intervals contain the true means,*



$$\Delta_R \le U_R = \frac{2b_R + \gamma\eta U_B}{1-\gamma+\gamma\eta}, \qquad U_B = \frac{2b_B}{1-\gamma}, \qquad \gamma = 0.9$$

FIGURE 3: Uncertainty propagation on state pairs. Both panels use the one-action kernel $P_\Lambda = \left(\begin{smallmatrix} 1-\eta & \eta \\ 0 & 1 \end{smallmatrix}\right)$ with $\gamma = 0.9$, $\eta = 0.20$, and $b_R = 0$. Solid arrows are pair transitions; dashed arrows show the backward contribution of future uncertainty. (a) An unresolved future with $b_B = 0.04$ gives $U_B = 0.800$ and $U_R \simeq 0.514$, so the bound does not certify the budget $w = 0.10$. (b) With the smaller future source $b_B = 0.004$, $U_B = 0.080$ and $U_R \simeq 0.051$, which certifies $\Delta_R \le 0.10$. The bars compare majorants on a common scale; they are neither distance intervals nor learning curves. The pair coupling is a mathematical construction, not an operation imposed on independent trajectories.

$\mathrm{TV}(P_{s,a}, \hat{P}_{s,a}) \le \beta^P(s,a)$, $\bar{D} \ge \|d_\sim\|_\infty$, *and* $D_{s,a} \ge \max_{x,y\in\Sigma_{s,a}} d_\sim(x,y)$ *for every row. Define, off the diagonal,*

$$\mathcal{T}_D^+[h] = \min\left\{\bar{D}, \max_a \left[(1-\gamma)\bar{g}_a + \gamma\hat{w}_a(h) + \pi_a\right]\right\},$$
$$\mathcal{T}_D^-[h] = \max_a \left\{(1-\gamma)\underline{g}_a + (\gamma\hat{w}_a(h) - \pi_a)_+\right\}. \qquad (32)$$

Both are monotone γ-contractions on $\mathfrak{C}$, and on $\mathcal{E}$ their fixed points satisfy $d_D^- \le d_\sim \le d_D^+$, with d_D^+ a pseudometric. If the reward intervals lie inside the Hoeffding ones, $\bar{D} \le R_{\max}$, and $\pi_a \le \gamma R_{\max} e_a^P$, then also $d^- \le d_D^-$ and $d_D^+ \le d^+$. Theorem 6 holds with the local source $2b_a$ replaced by $(1-\gamma)(\bar{g}_a - \underline{g}_a) + 2\pi_a$.

The constants can be computed from endpoints that are already valid on $\mathcal{E}$. Without knowledge of the supports, one takes $\Sigma_{s,a} = \mathcal{S}$ and $D_{s,a} = \bar{D} = \|d^+\|_\infty$ from the previous solution. When the support graph of the transitions is known, as in RiverSwim, gridworlds, and our ring models, $D_{s,a}$ is the largest upper endpoint among the known successors of the row, and $\beta^P(s,a)$ may use the row's own support size in (7), which vanishes for deterministic rows. Every refined upper endpoint is again valid on $\mathcal{E}$, so the refinement can be iterated until it stops changing. No union bound is added, and the anytime guarantee of Theorem 5 carries over unchanged. Section VIII-H measures how much is gained.

IV. FROM INTERVALS TO CERTIFIED REPRESENTATIONS

A. AN OPTIMISTIC COVER WITH A PACKING CERTIFICATE

Suppose the endpoints satisfy $\|\Delta\|_\infty \le w$, and fix a radius $r > w$. Starting from a single center, we repeatedly add any label whose d^+ distance to every existing center exceeds r.

Theorem 13 (Certified cover and anchor representation)**.** *The resulting center set C is an r-cover in d^+, and therefore in $d_\sim$. Its size satisfies*

$$|C| \le \mathcal{N}_\sim((r-w)/2; \mathcal{S}). \tag{33}$$

Define the representation

$$\phi_C(s) = (d^+(s,c))_{c\in C} \in \mathbb{R}^{|C|}. \tag{34}$$

Under the ℓ_∞ norm,

$$\kappa(\phi_C; \mathcal{S}) \le \max\{w, 2r\}. \tag{35}$$

If $C = \mathcal{S}$, then $\|\phi_C(s) - \phi_C(s')\|_\infty = d^+(s,s')$ and the distortion is at most w.

Proof. A center is added only if its d^+ distance to every previous center exceeds r. Since $d^+ - d_\sim \le w$, centers are more than $r - w$ apart in the true metric, and a true ball of radius $(r-w)/2$ contains at most one of them. This gives (33).

Because d^+ satisfies the triangle inequality, $|\phi_C(s)_c - \phi_C(s')_c| \le d^+(s,s')$ for every c. Choosing a center within r of s in d^+ and applying the reverse triangle inequality gives

$$d^+(s,s') - 2r \le \|\phi_C(s) - \phi_C(s')\|_\infty \le d^+(s,s'). \tag{36}$$

Combining this with $d_\sim \le d^+ \le d_\sim + w$ proves (35). When every label is an anchor, the coordinate $c = s$ attains $d^+(s,s')$ exactly. □

The theorem separates interaction from compression. The number of output coordinates adapts to the geometry, even though collecting the data still scales with the number of statistically distinct rows. For instance, $w \le \varepsilon/4$ and $r = \varepsilon/2$ give an $\varepsilon/2$-cover, distortion at most ε, and at most $\mathcal{N}_\sim(\varepsilon/8)$ anchor coordinates.

Figure 4 shows the construction on a concrete example. The three coordinates are distances to three centers, and the covering neighborhoods are ℓ_∞ cubes. No low-dimensional Euclidean isometry is involved; in fact, changing the norm changes the distances and can break the upper inequality in (36). The example has seven single-action self-loop states with $r(s_i) = i/6$, $\gamma = 0.9$, reward radii 0.04, and known transitions (zero transition radii). Its exact endpoints are $d^- = (d_\sim - 0.08)_+$ and $d^+ = \min\{1, d_\sim + 0.08\}$ off the diagonal.

B. EUCLIDEAN ENCODERS AND AN OBSERVABLE CERTIFICATE

Let $h_\phi(s,s') = \|\phi(s) - \phi(s')\|_2$. For any fitted encoder, the sandwich immediately gives

$$\kappa(\phi; \mathcal{S}) \le \left\|h_\phi - \bar{d}\right\|_\infty + \tfrac{1}{2}\|\Delta\|_\infty. \tag{37}$$

The bound holds no matter how the network was trained. The training residual, the confidence width, and a numerical margin together form a certificate that can be checked after the fact.

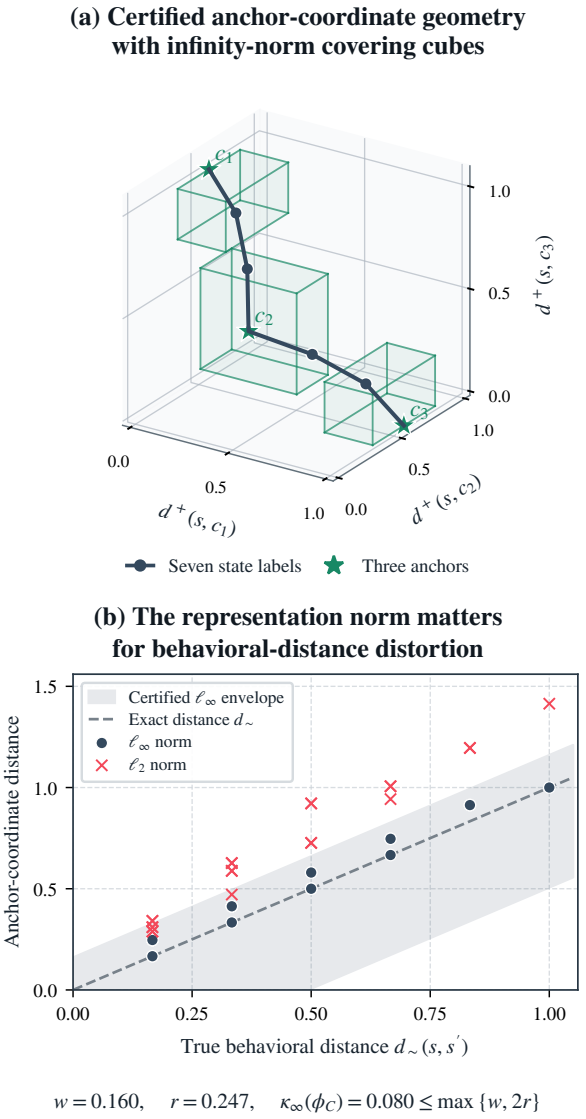


FIGURE 4: Certified anchor representations for the analytical self-loop example. (a) Coordinates from anchors $\{s_0, s_3, s_6\}$, with covering cubes restricted to $[0,1]^3$. Segments connect consecutive state labels. (b) All 21 unordered pairs under the infinity and Euclidean norms. The shaded envelope is $\max\{d_\sim - 2r, 0\} \le \|\phi_C(s) - \phi_C(s')\|_\infty \le d_\sim + w$, with $w = 0.160$, $r = 0.247$, and actual infinity-norm distortion 0.080. The envelope is a deterministic consequence of the theorem, not a statistical confidence band.

Proposition 14 (Approximation-aware fitting)**.** *If ϕ minimizes $\left\|h_\phi - \bar{d}\right\|_\infty$ over Φ up to an optimization error ξ, then*

$$\kappa(\phi; \mathcal{S}) \le \kappa_\Phi + \|\Delta\|_\infty + \xi. \tag{38}$$

The same bound holds when the encoder is fitted to the interval objective

$$\sup_{s,s'} \operatorname{dist}\left(h_\phi(s,s'), [d^-(s,s'), d^+(s,s')]\right).$$

Proof. For midpoint fitting, an encoder within $\kappa_\Phi + \zeta$ of the true metric is within $\kappa_\Phi + \zeta + \|\Delta\|_\infty/2$ of the midpoint; apply (37) and let $\zeta \downarrow 0$. For interval fitting, the same encoder has loss at most $\kappa_\Phi + \zeta$, because the true metric lies in every interval. A fitted distance outside an interval differs from any point inside it by at most its interval loss plus the interval width. □

A weighted squared objective $\mathcal{L} = S^{-2}\sum_{s,s'} v_{s,s'}(h_\phi - \bar{d})^2$ with $v_{s,s'} \geq v_{\min} > 0$ yields only

$$\kappa(\phi; \mathcal{S}) \leq S\sqrt{\mathcal{L}/v_{\min}} + \tfrac{1}{2}\|\Delta\|_\infty . \tag{39}$$

The factor S is the price of turning a mean-square residual into a uniform bound over finitely many pairs, so a small average loss does not by itself guarantee a small worst-pair distortion. If (37) is evaluated directly, however, no claim about the global optimality of the network is needed.

C. CONSISTENT TEMPORAL AND GLOBAL TARGETS

Temporal pairs and replay pairs should regress toward the same midpoint. A temperature that multiplies the temporal *loss weight* changes its emphasis while leaving its target intact. A temperature that multiplies the temporal *target*, by contrast, conflicts with the replay term whenever the same pair appears in both. If positive weights v_t and v_g regress one distance x toward τy and y respectively, the unconstrained optimum is

$$x^* = \frac{v_t\tau + v_g}{v_t + v_g} y, \tag{40}$$

which differs from y whenever $\tau \neq 1$. Common, unscaled targets are necessary for a zero-loss isometry, but they are not sufficient. The target must also be realizable by the encoder class, and a nonmetric midpoint or a non-Euclidean metric can rule this out.

V. CLOSING THE FINITE-STATE LEARNING LOOP

A. STRUCTURAL REACHABILITY WITHOUT A NAVIGATION ORACLE

Assumption 15 (Reset reachability)**.** The label set $\mathcal{S}$, the reset state s_0, and parameters $H \geq 1$ and $p \in (0, 1]$ are known. For every $s \in \mathcal{S}$ there exists a policy that visits s within $H - 1$ transitions from s_0 with probability at least p. The reward and transition models and the reaching policies are unknown. A reset is available after every episode.

The assumption asserts that reaching policies exist; it does not hand them to the agent. It is implied by a bound on expected hitting times: if every label can be hit in expected time at most L, Markov's inequality gives probability at least $1/2$ of hitting it within $2L$ transitions, so $H = \lceil 2L \rceil + 1$ and $p = 1/2$ suffice. Incremental controllability and reachability are different notions [25], [26], and we use the explicit horizon-probability condition above.

Fix a target distortion $\varepsilon > 0$ and allocate $\delta_m = \delta/2$ of the failure probability to model confidence. Define

$$\alpha_* = \min\left\{\frac{(1-\gamma)\varepsilon}{32R_{\max}}, \frac{p}{8H}\right\}. \tag{41}$$

Choose any integer $n_0 \geq 1$ such that

$$\sqrt{\frac{\Gamma\log 2 + \log(4SA\pi^2 n_0^2/(3\delta_m))}{2n_0}} \leq \alpha_*. \tag{42}$$

Such an integer exists and is found by increasing the quota until the inequality holds. It satisfies

$$n_0 = \widetilde{O}\left[\Gamma\max\left\{\frac{R_{\max}^2}{(1-\gamma)^2\varepsilon^2}, \frac{H^2}{p^2}\right\}\right], \tag{43}$$

where $\widetilde{O}$ hides logarithms in the problem parameters and the confidence level. The two terms play different roles and are best kept apart: the first resolves the metric, and the second makes navigation reliable.

B. OPTIMISTIC EXPLORATION OF UNRESOLVED ROWS

A row becomes known once its first n_0 samples have been retained, and $\mathcal{U} = \{(s,a) : n(s,a) < n_0\}$ denotes the set of unresolved rows. Each exploratory episode tries to execute one unresolved row and stops as soon as it succeeds. Propagated metric uncertainty assigns every unresolved row a bounded priority. At a snapshot, let

$$\begin{aligned}\chi_t(s,a) &= \max_{s'}\{2b_a(s,s') + \gamma\langle\Lambda^a_{s,s'}, U\rangle\},\\ v_t(s,a) &= \tfrac{1}{2} + \frac{\chi_t(s,a)}{2\chi_{\max}}, \qquad (s,a) \in \mathcal{U},\end{aligned} \tag{44}$$

where $\chi_{\max} = \max_{(x,b)\in\mathcal{U}}\chi_t(x,b)$, and set $v_t = 1$ if this maximum is zero. Priorities are frozen within an episode and may be refreshed less often to save computation. The proof uses only $1/2 \leq v_t \leq 1$, and the lower bound ensures that no row is starved of samples. The bounded majorant U_c of Proposition 7 may replace U in this rule. Neither choice affects the proof or the worst-case rate, and uniform priorities remain a valid baseline. Whether geometric priorities actually help is therefore an empirical question, and Section VIII takes it up.

The planner assigns a terminal payoff $v_t(s,a)$ to executing an unknown row and never queries that row's unknown transition. Known rows use their empirical transitions. The horizon-dependent recursion is

$$\begin{aligned}\hat{V}_0(s) &= 0,\\ \hat{Q}_h(s,a) &= \begin{cases} v_t(s,a), & (s,a) \in \mathcal{U},\\ \sum_x \hat{P}_{s,a}(x)\hat{V}_{h-1}(x), & (s,a) \notin \mathcal{U},\end{cases}\\ \hat{V}_h(s) &= \max_a \hat{Q}_h(s,a).\end{aligned} \tag{45}$$

The agent follows a maximizing action for its remaining horizon and breaks ties in favor of executing an unresolved row. The analysis admits any maximizing policy, but the tie-breaking rule is not innocuous. With equal payoffs, the recursion is indifferent to *when* a row is executed, and other rules can spend the slack of the horizon on known loops (Section VIII-G). After the first sample from an unknown row, or after H steps, the agent resets and replans. Samples from known rows encountered along the way may be logged, but the proof does not use them. Figure 5 summarizes the

Algorithm 1 Certified EBG with reset-based learned navigation

Input: Labels $\mathcal{S}$, actions $\mathcal{A}$, $s_0, H, p, \varepsilon, \delta$.

Select n_0 by (42); initialize every count to zero.
while some state-action count is below n_0 **do**
  Form $\mathcal{U}$ and empirical estimates from retained data.
  Refresh propagated metric priorities (44) at the chosen epoch frequency.
  Solve (45) with priorities frozen for this episode, breaking ties toward unresolved rows.
  Reset to s_0 and follow its horizon-dependent policy.
  if an unknown row is executed within H steps **then**
    Retain its reward and successor; increment its count.
  end if
end while
Compute $d^{\pm}$ by (12)–(13), reserving total numerical padding $\varepsilon/8$.
Expand endpoints by their residual margins and path-close the upper endpoint; call the results L, U.
Construct an $r = \varepsilon/2$ cover and its anchor map.
Plan empirical goal-reaching policies for the cover centers.

Output: Interval, cover, representation, and learned navigation policies.

resulting workflow: the acquisition loop of Algorithm 1, the outputs certified at its termination, and the order of the neural extension in Appendix F.

Lemma 16 (Learned exploration succeeds)**.** *Before termination, on any history whose known rows satisfy* $\mathrm{TV}(P, \hat{P}) \le \alpha_*$*, the policy defined by* (45) *executes an unknown row within* H *steps with conditional probability at least* $p/4$*.*

Proof. For any unresolved row (g, a), some true policy reaches g within $H-1$ steps with probability at least p and then executes a. Stopping earlier at another unresolved row yields payoff at least $1/2$, so the optimal true expected terminal payoff is at least $p/2$. For any fixed stopped policy, coupling the true and empirical transitions on its known rows shows that its true and empirical expected payoffs differ by at most $H\alpha_*$. Empirical optimization loses this amount once when the true optimizer is evaluated and once more when the empirical optimizer is executed, so the executed policy has true expected payoff at least $p/2 - 2H\alpha_* \ge p/4$. Terminal payoffs are at most one, hence the probability of executing an unresolved row is at least this payoff. □

Theorem 17 (Interaction-only closed-loop certificate)**.** *Under Assumption 15, with probability at least* $1-\delta$*, Algorithm 1 terminates within*

$$T \le H \left\lceil \frac{8}{p} \left(SAn_0 + \log \frac{2}{\delta} \right) \right\rceil \tag{46}$$

interaction steps. On the same event:

1) *the returned endpoints satisfy* $L \le d_\sim \le U$ *and* $\|U - L\|_\infty \le \varepsilon/4$*;*
2) *the visited centers form an* $\varepsilon/2$*-cover, with* $|C| \le \mathcal{N}_\sim(\varepsilon/8; \mathcal{S})$*;*
3) *the constructed* ℓ_∞ *representation has* $\kappa(\phi_C; \mathcal{S}) \le \varepsilon$*, using* U *as its anchor metric;*
4) *for each center, its learned empirical policy reaches that center within* $H-1$ *steps with true probability at least* $p/2$*.*

If a Euclidean encoder is fitted as in Proposition 14, its distortion is at most $\kappa_\Phi + \varepsilon/4 + \xi$*.*

The proof (Appendix C) combines Lemma 16 with a success-count bound for adapted Bernoulli trials. Observe that the exploration planner uses neither replay coverage nor a trained representation. Geometry enters only through the bounded priorities, while the labels and the row confidence carry the statistical guarantee. Since every row holds n_0 retained observations at termination, balanced state and pair replay can be formed, with marginal masses $1/S$ and pair masses $1/S^2$. These distributions are earned by the algorithm and make no claim about the occupancy of a task policy.

Combining (43) with (46) gives

$$T = \widetilde{O}\left[\frac{H}{p} SA\Gamma \max\left\{ \frac{R_{\max}^2}{(1-\gamma)^2\varepsilon^2}, \frac{H^2}{p^2} \right\} \right]. \tag{47}$$

The factor H/p is the cost of navigation, and the inverse-square factor is the cost of identification. The covering bound compresses the output, but it does not replace S in the interaction bound. We stress that the theorem concerns statistical efficiency only; solving all pairwise transport problems, or optimizing a neural fitting objective globally, can still be computationally expensive. The smallest admissible quota can be found by doubling followed by binary search on (42). A budget computed with different radii, or without this quota, is not the value of (46), and throughout the evaluation we keep prescribed budgets separate from observed interaction counts.

C. INFLUENCE PRIORITIES AND TARGETED QUERIES

The score (44) looks forward. It ranks a row by the uncertainty that its pairs carry, including what they inherit from their futures. Sampling a row, however, reduces only that row's radii, and the effect of the reduction travels backwards along coupled futures. Corollary 8 makes this backward direction explicit. For the maximizing pair policy σ of the majorant recursion and a nonnegative weight w on pairs,

$$w^\top U = 2\rho^\top b^\sigma, \qquad \rho = \left(I - \gamma (P_\Lambda^\sigma)^\top\right)^{-1} w, \tag{48}$$

so that ρ is the discounted occupancy of the pair process started from w. Half of the source $b^\sigma(s,s')$ belongs to each of the two rows it involves, and the share of $w^\top U$ owed to row (s,a) is therefore

$$\iota_w(s,a) = \sum_{s' :\, \sigma(s,s') = a} \{\rho(s,s') + \rho(s',s)\}\, b^\sigma(s,s'), \tag{49}$$

with the shares of all rows summing to $w^\top U$. Since radii scale as $n^{-1/2}$, the score $\iota_w(s,a)/\{n(s,a)+1\}$ approximates the reduction obtained per additional sample. We call these

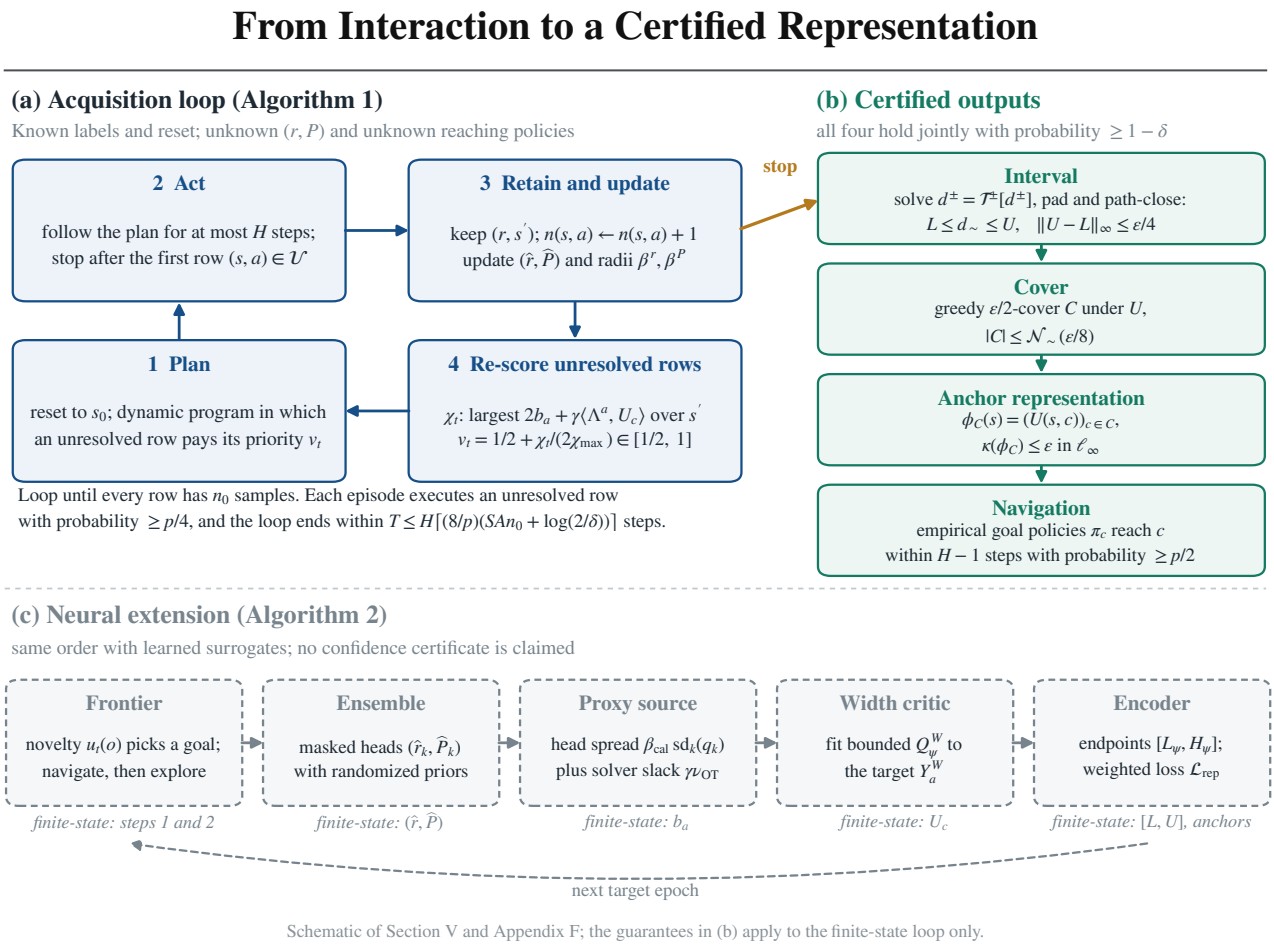


FIGURE 5: Workflow of certified epistemic bisimulation geometry. (a) Acquisition loop of Algorithm 1: each episode plans toward an unresolved row with priorities in $[1/2, 1]$, retains one sample, and refreshes the confidence quantities. (b) Outputs computed once every row has n_0 samples; by Theorem 17, they hold jointly with probability at least $1 - \delta$. (c) The neural extension of Algorithm 2 follows the same order with learned surrogates and carries no confidence certificate. The diagram is schematic and contains no measurement.

adjoint priorities and map them to $[1/2, 1]$ as in (44), so Theorem 17 continues to apply. A uniform w targets the mean width. A weight on a single pair targets one decision, for example whether $d^-(s, s') > 0$ certifies that s and s' are not bisimilar. Rows with no influence on that pair can then be retired early, which the full certificate of Theorem 17 does not permit. Section VIII-J evaluates both uses.

D. WHAT AN EXACT ABSTRACTION CAN ADD

Corollary 18 (Certified quotient composition)**.** *Suppose a fixed decoder f yields M labels and is an exact MDP homomorphism: under every action, the mean rewards and the distributions of $f(S_{t+1})$ coincide for all raw states with the same decoded label. Assume reset reachability on these labels. Then Theorem 17 applies with $S = M$ and the support bound of the quotient, and its representation and interval pull back through f. If a separate procedure certifies such a decoder with C_{dec} interactions and confidence $1 - \delta/2$, the combined interaction bound is C_{dec} plus (46) run with failure budget $\delta/2$.*

The proof is exact lumpability. Decoded successor samples share one conditional law, and bounded rewards share a conditional mean, although their noise distributions need not coincide. The lifted quotient metric is a fixed point of the raw operator. Approximate ε-nets, by contrast, are not in general exact homomorphisms: averaging over the states of a cell can change its transition law, so an approximate abstraction needs its own aggregation error and a controlled sampling rule.

When observations $o \sim q(\cdot \mid s)$ are drawn from disjoint emission supports, a decoder exists. Its existence, however, neither makes it free to learn nor certifies its behavior on unseen observations. The rich-observation algorithm of Appendix F should be read as a practical proposal; a bound in terms of $\mathcal{N}_\sim$ alone would additionally require a decoder, or a guarantee of statistical sharing of the kind made explicit here.

VI. STATISTICAL LIMITS AND ROBUSTNESS

A. A SMALL TRUE METRIC DOES NOT IMPLY CHEAP IDENTIFICATION

One might expect that a geometry with a small covering number can be learned with correspondingly few samples. The next result shows that, without further structure, this expectation is wrong.

Theorem 19 (Obstruction to a geometry-only sample bound)**.** *Let $R_{\max} = 1$, $0 < \varepsilon \le 1/16$, and $0 < \delta < 1/4$. There is a finite MDP with $S + 1$ labels, A actions, and identically zero true metric such that any algorithm that is uniformly δ-correct over the family constructed below, and returns simultaneous metric intervals of width at most ε, requires on that MDP*

$$\mathbb{E}T \ge \frac{SA}{64\varepsilon^2}\,\mathrm{kl}(1-\delta,\delta). \tag{50}$$

The bound holds even with generative access to every state-action pair. Hence $\mathcal{N}_\sim(\varepsilon) = 1$ and $D = 0$ on the hard instance do not yield a bound independent of the statistical state complexity.

Here kl denotes binary relative entropy. The construction is simple. Every state is absorbing. A reference label has known Bernoulli mean $1/2$ under every action, and in the base instance so do all other labels. For each nonreference row (i, a), one alternative changes only that row's mean, to $1/2 + 4\varepsilon$. For absorbing self-loops,

$$d_\sim(i,j) = \max_a |r(i,a) - r(j,a)|, \tag{51}$$

independently of γ. An interval of width ε at the pair $(i, \mathrm{reference})$ cannot contain both 0 and 4ε, so an adaptive change-of-measure argument [34] forces an inverse-square number of samples from every row that can be perturbed independently. Appendix D gives the constants.

The theorem does not rule out intrinsic rates under a shared parametric model, a certified decoder, or known smoothness. What it rules out is deriving such rates from the geometry of the true metric alone. In particular, observing similar rewards does not by itself license pooling rows that have not been observed.

B. TWO CLOCKS ON A CONTROLLED GEOMETRIC FAMILY

On a family whose geometry we control, the cost of finding a cover and the cost of certifying it can be computed exactly, and they differ by two powers of the accuracy. As in the closed loop, a label may serve as a center only after it has been queried at least once. We write T_{cover} for the number of generative queries needed before the queried labels contain an ε-cover of the true metric, and T_{resolve} for the number needed to return simultaneous intervals of width at most ε at every pair.

Theorem 20 (Coverage and resolution separation)**.** *For each integer $d \ge 1$ and $m \ge 2$, there is a known-dynamics family with $A = d + 1$ actions, $M = m^d$ grid labels, and one reference label. Let $\varepsilon = 1/(32m)$. The grid metric is exactly a d-dimensional ℓ_∞ lattice metric, the doubling dimension is $O(d)$, and $\mathcal{N}_\sim(\varepsilon) = M + 1 = \Theta(\varepsilon^{-d})$. With generative access,*

$$T_{\mathrm{cover}} = \Theta(M), \tag{52}$$

$$T_{\mathrm{resolve}} = \widetilde{\Theta}(M\varepsilon^{-2}) = \widetilde{\Theta}(\varepsilon^{-(d+2)}) \tag{53}$$

for any fixed confidence bounded away from one half. For $0 < \delta < 1/2$, resolution at confidence $1 - \delta$ has an upper bound $O(M\varepsilon^{-2}\log(M/\delta))$ and a lower bound $\Omega(M\varepsilon^{-2}\,\mathrm{kl}(1-\delta,\delta))$.

All labels self-loop. At grid point $q \in \{0, \dots, m-1\}^d$, the first d actions have known deterministic means

$$r(q,j) = \frac{1}{4} + \frac{q_j}{4m}, \qquad j = 1, \dots, d. \tag{54}$$

The last action has Bernoulli mean $\mu_q = 3/4 + 2\varepsilon\theta_q$ with $\theta_q \in \{-1, 1\}$, and every reference reward is zero. Distinct grid points are separated by at least $1/(4m) = 8\varepsilon$ in the known coordinates, while the unknown action separates them by at most 4ε. Therefore

$$d_\sim(q,q') = \frac{\|q - q'\|_\infty}{4m},$$
$$d_\sim(q, \mathrm{reference}) = \mu_q. \tag{55}$$

Distinct labels are more than ε apart, so every ε-cover contains every label, and a cover is found as soon as each label has been queried once, at a cost of $\Theta(M)$. Resolving the distance to the reference is another matter: it reveals each independent sign θ_q, and that costs an inverse-square number of samples per label. The full proof also exhibits an interval estimator that attains the upper bound.

The two extra powers are a sharp property of this family; we do not claim them as a universal law for continuous dynamics or rich observations. A fixed finite grid has a saturated covering number at fine scales, the exponents refer to a family indexed by resolution, and we do not claim that the coordinate dimension d equals the exact doubling dimension of each finite space.

For continuous successor distributions, empirical transport converges at rates that can depend on dimension, rather than at a universal $n^{-1/2}$ rate [35], [36]. Likewise, a statistically learned ground metric can enter a transport confidence radius only with uniform control of its dependence on the data. An optimistic cover is geometrically safe, but it does not by itself supply such a concentration bound.

C. REWARD GEOMETRY AND GOAL REACHABILITY ARE DIFFERENT

Proposition 21 (Reachability is not identified by extrinsic bisimulation)**.** *If every reward is zero, then $d_\sim$ is identically zero for every transition kernel, yet two such kernels can reach a specified label from s_0 with different probabilities.*

The zero cost is a fixed point of (3). One kernel may move s_0 to a target with probability one while another keeps it at s_0

forever, and both have the same zero metric. A certificate for the extrinsic-reward metric therefore cannot take the place of the reachability assumption or of a learned navigation model.

A goal-sensitive alternative adds a declared family $\mathcal{G}$ of bounded goal rewards:

$$\mathcal{T}_{\mathcal{G}}[h](s,s') = \max_a\{(1-\gamma)\max_{g\in\mathcal{G}}|r_g(s,a)-r_g(s',a)| + \gamma\,\mathrm{OT}_h(P_{s,a},P_{s',a})\}. \tag{56}$$

This operator is still a γ-contraction, and the confidence arguments extend whenever each goal reward has a valid confidence radius. Its geometry and covering numbers, however, are those of the goal family, and they should not be reported as the extrinsic bisimulation geometry.

D. EXOGENOUS INVARIANCE

Theorem 22 (Exact exogenous invariance)**.** *Suppose the state is (x,e) and*

$$P(x',e' \mid x,e,a) = P_x(x' \mid x,a)P_e(e' \mid e),\quad r((x,e),a) = r_x(x,a). \tag{57}$$

Then, for all x,y,e,e',

$$d_\sim((x,e),(y,e')) = d_x(x,y), \tag{58}$$

where d_x is the endogenous bisimulation metric. Consequently the true metric quotient and its covering numbers do not depend on the cardinality of the exogenous space.

Proof. Lift d_x to a cost that depends only on (x,y). Every coupling of the full successor laws projects to an endogenous coupling of the same cost, which gives a lower bound by endogenous transport. Conversely, an optimal endogenous coupling lifts to a full coupling by coupling the exogenous marginals independently, which gives the reverse bound. The lifted cost is therefore a fixed point of (3), and uniqueness proves the claim. □

The invariance is geometric rather than statistical. A learner that estimates a separate model for every raw state (x,e) still pays for the exogenous space, as Theorem 19 shows. Statistical invariance requires a valid endogenous abstraction, whether learned or given, in the spirit of exogenous-state filtering [29].

E. ALEATORIC NOISE AND EXHAUSTED REGIONS

Proposition 23 (Vanishing epistemic width)**.** *In a finite stationary model, if every row in K is sampled without bound and $\eta_K \to 0$, then the confidence width on $K \times K$ converges to zero on the simultaneous confidence event. For $K = \mathcal{S}$ and fixed Γ,*

$$\|\Delta\|_\infty = O\left(\frac{R_{\max}}{1-\gamma}\sqrt{\frac{\Gamma+\log(SAn/\delta)}{n}}\right) \tag{59}$$

when every row has at least n samples and the radii are evaluated at those retained counts.

The proposition follows from Proposition 9 and (7). For the displayed rate one may retain exactly n samples per row, which avoids any dependence on unequal counts.

Consider a noisy television whose screen states share a reward and a transition law and have no downstream effect on control. These screen states are bisimilar. A Bayes-optimal squared-error predictor of the next screen observation incurs the irreducible loss $\mathrm{tr}\,\mathrm{Cov}(O_{\mathrm{next}})$, which stays positive after the screen distribution has been learned [37]. The metric confidence width, in contrast, vanishes once the finite behavioral model is resolved. The constant in (59) can still depend on the successor support and on the complexity of decoding, so in general it is not independent of the number of screen states.

Vanishing width does not mean that a task policy spends no time at the television, since navigation, extrinsic rewards, and the bonus schedule also determine where the agent goes. Width decay is a property of estimation; whether the agent actually avoids noisy observations would have to be established by a separate visitation experiment.

VII. TOWARD RICH OBSERVATIONS

The certificates of Sections III–V require finite labels and a confidence radius for every row, and rich observations remove both. It helps to separate what transfers from what does not. Three ingredients carry over unchanged. The bounded operators apply to any learned model whose rows carry a stated error; Proposition 10 prices any approximate solver, neural or otherwise; and (37) turns any fitted encoder into a distortion certificate that can be checked after training. What does not carry over is the simultaneous model confidence of Lemma 2. Deep ensembles, Monte Carlo dropout, and temperature calibration give practical estimates of aleatoric and epistemic uncertainty for decision making [38], but these estimates are calibrated on held-out data and are not guaranteed uniformly over unseen observations. Their characteristic failure is collapse: unvisited observations are mapped onto familiar embeddings, and all heads then agree, wrongly. A certified decoder would restore the guarantee through Corollary 18.

Appendix F gives a neural instantiation that preserves the structure of the finite construction. It consists of an ensemble of bounded reward and stochastic successor heads, a bounded symmetric width critic trained on the pair-space target of Proposition 7, explicit surrogate endpoints, a normalized representation loss that regresses temporal and replay pairs onto the same frozen target, and anchor-based frontier navigation (Algorithm 2). Section VIII-K evaluates its representation and width components on a block MDP with a strong exogenous distractor, where the true metric is known. That study measures distortion, invariance, and the empirical coverage of the surrogate intervals, but it does not certify them, and image-based tasks with learned policies are left for future work.

VIII. FINITE-STATE EVALUATION

The theory makes three kinds of claims that can be tested: that the intervals are valid, that they are sharp enough to be useful,

TABLE 2: The eleven studies: question (recorded endpoint) and access model.

Study	Question (endpoint)	Access
V1	Are the intervals valid and the anchor map accurate? (all-pair coverage)	Generative rows
V2	Does the majorant localize inherited width? (pair width, majorant)	Partially known MDP
V3	Navigation or priorities: which sets acquisition cost? (steps to a row quota)	Reset interaction
V4	How large are the two-clock budgets? (prescribed queries)	Formula only
V5	What does a certificate cost against other intervals? (coverage, width)	V1 samples
V6	Do coverage, width, and compression persist up to $S = 128$?	Generative rows
V7	Do priorities improve the anytime certificate? (width at step budgets)	Reset interaction
V8	How much do refinements narrow the intervals, and which decisions are certified?	V1 and V6 samples
V9	Does the picture hold on RiverSwim? (coverage, width, navigation)	Generative, reset
V10	Do adjoint priorities help, overall or for one query?	Reset interaction
V11	What does the neural instantiation achieve? (distortion, invariance)	Block MDP

and that the closed loop acquires its data at a reasonable cost. Table 2 lists the eleven studies we ran to test them, the question each answers, and its access model. V4 evaluates the explicit formula of a theorem rather than observed stopping times; all other studies are sampling experiments. Every configuration was fixed in a written specification before execution, with two post hoc exceptions, namely the tie-breaking rerun of Section VIII-G and the target-bias computation of Section VIII-K. The pilot that set the V7 quota used one seed that also appears among the reported V7 runs, whereas the pilots that fixed the designs of V9 and V10 used seeds excluded from the reported ones. Appendix E gives seeds, sampling schemes, solver tolerances, and statistical summaries, and the public repository [8] contains the code, every specification and run manifest, and all seed-level records.

A. V1: CALIBRATION AND CERTIFIED REPRESENTATIONS

We use two ring MDPs, with (S, A, Γ, γ) equal to $(6, 2, 2, 0.6)$ and $(8, 2, 2, 0.9)$, $R_{\max} = 1$, and $\delta = 0.1$. Action 0 moves counterclockwise and action 1 clockwise, each with probability 0.75; otherwise the state is unchanged. For label i the mean rewards are

$$r(i,0) = 0.08 + \frac{0.74i}{S-1},$$
$$r(i,1) = 0.12 + \frac{0.68(3i \bmod S)}{S-1}. \tag{60}$$

Reward observations are Bernoulli and, given the row, independent of the successor draw. Each model is run with 500 sampling seeds, each using nested prefixes of $0, 128, 512, 2048, 8192, 32768, 131072$ samples per row; no navigation is involved. At each checkpoint we record

$$e_t = \|\bar{d}_t - d_\sim\|_\infty, \qquad w_t = \|d_t^+ - d_t^-\|_\infty,$$
$$I_t = \mathbf{1}\{d_t^- \le d_\sim \le d_t^+ \text{ at all pairs}\}, \tag{61}$$

where $d_\sim$ is the metric of the exact model and $d_t^\pm$ are the empirical endpoints with residual padding. Coverage is checked at every pair, including the diagonal, and a run counts as covered only if $I_t = 1$ at all seven checkpoints; the indicator is not conditioned on the confidence event.

All 500 runs of each model were covered at every checkpoint. The two-sided 95% Clopper–Pearson interval for the probability that a run is covered at all checkpoints is therefore $[0.9926, 1]$, a statement that of course says nothing about times we did not record. At $n = 0$ the endpoints are exactly 0 and 1 off the diagonal and $e_0 \le 0.5$. Every covered interval satisfies $e_t \le w_t/2$, and violations of the majorant inequality and of the triangle inequality for the upper endpoint stay below the numerical tolerance of Appendix E. The intervals are valid, but they are also conservative. At the final checkpoint the mean width is 0.109 against a mean midpoint error of 0.0016 for the low-discount model, and 0.397 against 0.0080 for the high-discount model. Since S and γ change together between the two models, the difference cannot be attributed to the discount alone.

We also audited the anchor representation. When the greedy cover with $r = 0.25$ is applied to every final low-discount endpoint matrix, each observed width is below r, so Theorem 13 applies. The mean number of anchors is 6.00, and the largest infinity-norm distortion over all seeds is 0.057, comfortably below the bound $\max\{w, 2r\} = 0.5$. The audit concerns the anchor map alone, with no Euclidean approximation term. Since all six labels remain anchors at this radius, it tests accuracy rather than dimensional compression; V6 supplies the latter.

B. V2: INHERITED WIDTH AND MAJORANT SHARPNESS

Four labels form a current pair $(0, 1)$ and a future pair $(2, 3)$. There is one action, $\gamma = 0.8$, and the rewards at labels 0 and 1 are known to be 0.1. Transitions are known: each current label stays with probability $1 - \eta$ and moves to its corresponding future label with probability η, and future labels self-loop. The future labels have unknown Bernoulli rewards with means 0.3 and 0.7, and only these two reward rows are sampled. We use $\eta \in \{0.02, 0.10, 0.30\}$, 100 seeds, and nested counts $0, 32, 128, 512, 2048, 8192$ per uncertain row. The current pair therefore has zero local source, and all transition radii vanish, so any width at the current pair is inherited from its future. Figure 7 compares the actual width with the majorant U_c and with the global bound $\min\{1, 2\|b\|_\infty/(1-\gamma)\}$; the source-only value, zero, is a negative control and not a valid bound on the width of this pair.

Every current-pair interval covered the reference distance at every checkpoint. At $\eta = 0.02$ and $n = 8192$, the global bound is 13.50 times larger than the mean pair majorant; in other words, the majorant localizes the uncertainty to the

TABLE 3: Interval widths on the V1 samples, means over 500 seeds per model; n is samples per row. The columns give the mean midpoint error e_t of the baseline certificate; the mean supremum width of the baseline certificate (V1), of the diameter-refined ($\bar{D}$) and support-refined certificates of Proposition 12 (V8), and of the simulation-lemma interval (V5); and the all-pair coverage (Cov.) and width of the pointwise and simultaneous bootstrap intervals (V5). Every certified and simulation-lemma interval covered the true metric at every pair in every run. The bootstrap intervals have nominal level 0.9, no finite-sample guarantee, and were not computed at $n = 0$.

				Certified width			Sim.-lemma	Bootstrap, pointwise		Bootstrap band	
S	γ	n	e_t	Base	$\bar{D}$	Support	width	Cov.	Width	Cov.	Width
6	0.6	0	0.194	1.000	1.000	1.000	–	–	–	–	–
6	0.6	128	0.194	1.000	1.000	1.000	1.000	0.342	0.105	0.912	0.145
6	0.6	2,048	0.062	0.682	0.594	0.491	0.756	0.374	0.027	0.890	0.037
6	0.6	8,192	0.007	0.396	0.291	0.240	0.400	0.368	0.014	0.928	0.018
6	0.6	131,072	0.002	0.109	0.074	0.062	0.110	0.354	0.003	0.920	0.005
8	0.9	0	0.317	1.000	1.000	1.000	–	–	–	–	–
8	0.9	128	0.317	1.000	1.000	1.000	1.000	0.476	0.077	0.918	0.097
8	0.9	2,048	0.225	0.995	0.994	0.984	1.000	0.450	0.020	0.910	0.024
8	0.9	8,192	0.154	0.966	0.966	0.610	1.000	0.452	0.010	0.874	0.012
8	0.9	131,072	0.008	0.397	0.213	0.116	0.445	0.454	0.002	0.886	0.003

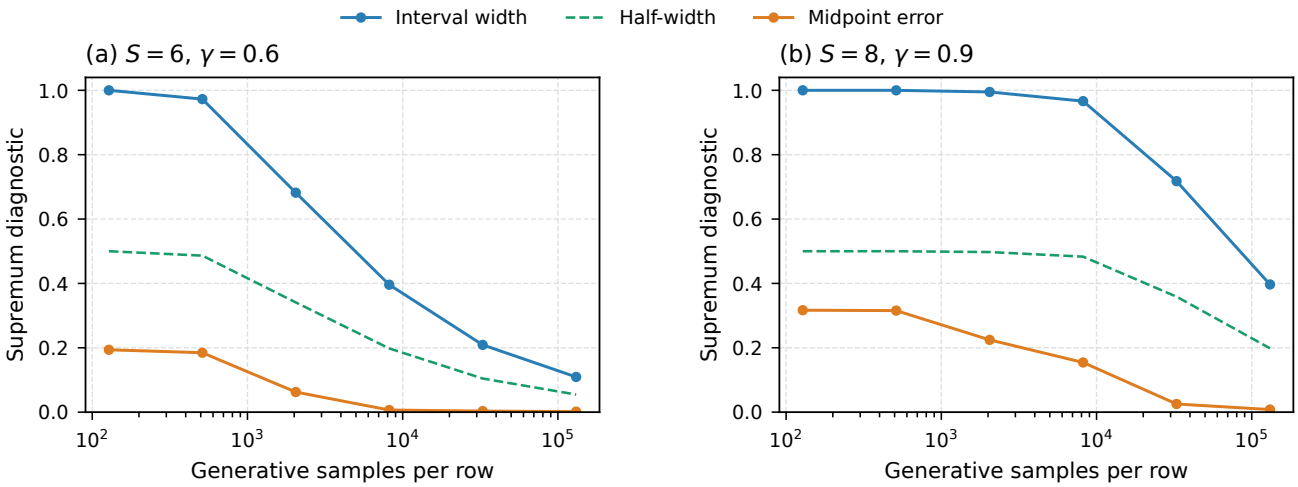


FIGURE 6: V1: interval width, midpoint error, and half-width against the number of samples per row (logarithmic axis), averaged over 500 sampling seeds. The 95% seed-bootstrap intervals of these means are drawn as shading but are narrower than the lines at most checkpoints; they are not behavioral-distance intervals. The zero-sample checkpoint is reported in Table 3. The two panels differ in both state count and discount.

pair that actually inherits it. This gain is a property of the family, and it shrinks as the escape probability grows. From 512 samples per row onward the majorant coincides with the actual width in every run, while at smaller counts the clipping of the endpoints separates the two. We should not read too much into this agreement. On a one-action chain the coupling is forced, so the experiment shows that the recursion can attain its bound, not that it is tight in general.

C. V3: A MATCHED NAVIGATION/PRIORITY ABLATION

We cross two priority rules with two navigation rules, random and optimistic, on two resettable MDPs with 32 seeds each. Source-only priorities use $\max_{s'} 2b_a(s, s')$, propagated priorities add the coupling expectation of U_c, both are mapped to $[1/2, 1]$, and both are refreshed every 128 episodes. The six-state corridor has one action that returns to label 0 and one that advances by one label with probability 0.85 and otherwise stays. The seven-state branching task is a binary tree of depth two: internal actions reach their designated child with probability 0.8 and otherwise stay, and both leaf actions return to the root. Episode horizons are 9 and 7; both tasks use $\gamma = 0.6$, $R_{\max} = 1$, $\Gamma = 2$, and $\delta = 0.1$, with reward means in Appendix E. A run ends when every row holds 64 retained samples, within a budget of 250000 steps. This diagnostic quota is not the n_0 of (42), so completion does not imply the

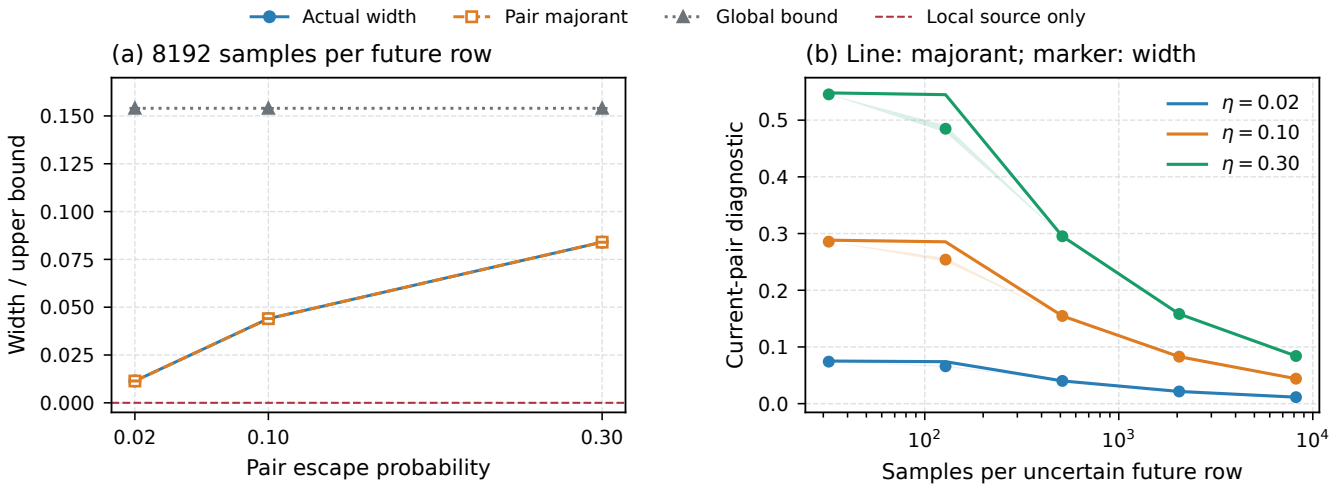


FIGURE 7: V2: uncertainty inherited from coupled futures in a partially known four-state MDP. (a) Bounds and actual width against escape probability at 8192 reward samples per uncertain row. (b) Pair majorants and actual widths against sample count for the three escape probabilities. Markers aggregate 100 seeds, and the intervals describe the sampling variation of their means. The known current rows have zero local source. The zero-sample checkpoint cannot be drawn on the logarithmic axis of (b) and is omitted.

metric-accuracy event of the closed-loop theorem. For a given task and seed, every variant sees the same first 64 reward and successor outcomes of each row, so completed variants return identical empirical models and differ only in how they acquired them.

The random controller serves as a negative control. For random navigation, priority changes have no effect by construction. On the corridor, the paired change in mean steps (propagated minus source-only) is 0.000 with a 95% seed-bootstrap interval $[0.000, 0.000]$. On the branching-return task, the paired change in mean steps (propagated minus source-only) is 0.125 with a 95% seed-bootstrap interval $[0.000, 0.375]$. These tasks do not demonstrate a systematic acquisition advantage from propagated priorities. Navigation, by contrast, matters exactly where the task makes it matter. Under source-only priorities, random navigation needs 11.09 times as many steps as optimistic navigation on the corridor, where rows deep in the chain are hard to reach, but only 1.03 times as many on the shallow branching task.

D. V4: EXPLICIT CONFIDENCE BUDGETS

For the controlled family of Theorem 20 we evaluate the prescribed quota (74) at $\delta = 0.1$, $d \in \{1, 2, 3\}$, and $m \in \{4, 6, 8, 10, 12\}$, keeping reference visits, integer rounding, and confidence logarithms (Table 5). The budgets are $M + 1$ for visiting every label and $Mn + 1$ for the fixed-quota interval estimator. The gap is striking even at moderate sizes: at $d = 2$ and $m = 12$, finding the cover takes 145 queries, whereas certifying it at this confidence takes about 1.35×10^9. The extra inverse-square factor comes from identifying independent means in a generative self-loop family. It should not be mistaken for a universal scaling law for interactive RL, nor for a matching lower bound on all the factors H, p, Γ, γ in (47).

E. V5: THE PRICE OF A CERTIFICATE

A certificate is only worth its width if cheaper alternatives are not equally reliable, so V5 compares four intervals computed from the same V1 samples (Table 3). Besides the certified endpoints, we form a coarser certified interval around the plug-in metric $\hat{d}$ of the empirical model, $[\hat{d} - \rho, \hat{d} + \rho]$ with $\rho = \|b\|_\infty / (1 - \gamma)$. This simulation-lemma interval omits pair-space propagation and is valid on the same event (Appendix E). We also form two uncertified intervals by the nonparametric bootstrap [39]: pointwise percentile intervals, and a simultaneous band whose half-width is the 90th percentile of the largest deviation of a replicate metric from $\hat{d}$. Both have nominal level $1 - \delta = 0.9$.

Both certified intervals covered the true metric in every run and at every checkpoint. Because every row holds the same number of samples here, the pair-space operators are at most 10% (low discount) and 15% (high discount) narrower than the simulation-lemma interval; their advantage only shows when local sources differ across pairs, as they do in V2. The pointwise bootstrap intervals covered all pairs simultaneously in only 0.328–0.374 of the low-discount runs and 0.450–0.476 of the high-discount runs, which is to be expected, since a pointwise level does not control 15 or 28 pairs jointly. The simultaneous band behaved better. It stayed close to its nominal level at each checkpoint, with coverage of at least 0.882 and 0.874, and at the final checkpoint it was 23.7 and 130.9 times narrower than the certified interval. Across all six checkpoints, however, only 297 and 292 of the 500 band runs covered at every checkpoint, against 500 of 500 certified

TABLE 4: V3: interaction steps to a quota of 64 samples per row, with 95% seed-bootstrap intervals; all 32 runs of every variant completed. "Local" is a source-only priority, not a source-only confidence certificate.

Task	Navigation	Local priorities	Propagated priorities
Corridor	Random	39,458.1 [38,356.6, 40,591.8]	39,458.1 [38,356.6, 40,591.8]
Corridor	Optimistic	3,559.6 [3,553.7, 3,565.8]	3,559.6 [3,553.7, 3,565.8]
Branching return	Random	2,906.1 [2,851.5, 2,961.6]	2,906.1 [2,851.5, 2,961.6]
Branching return	Optimistic	2,811.3 [2,805.6, 2,817.2]	2,811.4 [2,805.8, 2,817.3]

TABLE 5: V4: calculated budgets from the controlled-family theorem, with $M = m^d$, $\varepsilon = 1/(32m)$, and $n = \lceil 8\varepsilon^{-2}\log(2M/0.1)\rceil$ samples per row. The cover budget $M+1$ grows as m^d; the prescribed resolution budget $Mn+1$ grows as m^{d+2} up to the logarithm in n. These are formula evaluations, not observed stopping times or fitted scaling laws, and carry no confidence bands; $Mn+1$ is rounded to three significant digits.

		$d=1$			$d=2$			$d=3$		
m	ε^{-1}	$M+1$	n	$Mn+1$	$M+1$	n	$Mn+1$	$M+1$	n	$Mn+1$
4	128	5	574361	2.30×10^6	17	756066	1.21×10^7	65	937770	6.00×10^7
6	192	7	1411889	8.47×10^6	37	1940301	6.99×10^7	217	2468712	5.33×10^8
8	256	9	2660853	2.13×10^7	65	3751079	2.40×10^8	513	4841306	2.48×10^9
10	320	11	4340382	4.34×10^7	101	6226660	6.23×10^8	1001	8112938	8.11×10^9
12	384	13	6465225	7.76×10^7	145	9396540	1.35×10^9	1729	12327856	2.13×10^{10}

runs. The validity of the band is asymptotic and holds one snapshot at a time. Section IX returns to what the remaining gap consists of.

F. V6: SCALE AND COMPRESSION

V6 asks whether the behavior of V1 persists as the number of labels grows, and whether the anchor map compresses when the geometry allows it. We use exactly lumpable random MDPs with $S \in \{16, 32, 64, 128\}$ labels, two actions, two successors per row, and $\gamma = 0.6$, in the spirit of Garnet problems [40]. The labels form four latent blocks that share their reward means and block transition laws, but each label draws its own successor labels, so the raw transition matrices differ while the true metric has exactly four distinct points, with smallest nonzero distance 0.259. Each size uses 100 seeds and nested prefixes of 2048 to 131072 samples per row.

All 400 of 400 baseline runs covered the true metric at every pair and checkpoint. At 131072 samples per row the mean width grows only from 0.103 at $S = 16$ to 0.108 at $S = 128$, in line with the logarithmic dependence of the radii on SA in (59). With $r = 0.2$, the greedy cover returned exactly four anchors in every run of every size. Thus 128 labels are represented by four ℓ_∞ coordinates, with distortion far below the bound $\max\{w, 2r\} = 0.4$, a compression that V1 was too small to exhibit. Midpoint errors are larger than in V1, 0.031 at $S = 128$, for a simple reason: at a pair with zero true distance the lower endpoint is clipped at zero, so the midpoint sits at half the upper endpoint. One certificate took a median of 1.4 s at $S = 128$ on a shared workstation, so computation is not the bottleneck at this scale.

G. V7: PRIORITIES AND THE ANYTIME CERTIFICATE

V3 measured the cost of completing a small quota, an endpoint on which the order of acquisition hardly matters. Priorities, however, are meant to change which rows are resolved first, so V7 instead measures the certificate available at fixed interaction budgets. On the two V3 tasks with optimistic navigation, every row must hold 4096 retained samples, and we compare uniform priorities ($v_t = 1$, the R-MAX choice), source-only priorities, and propagated priorities on 32 matched seeds. At 5000 to 80000 interaction steps we compute the snapshot endpoints from all retained data and record their mean pairwise width; the supremum width remains one at these budgets, because some row is still unsampled or nearly so.

With the specified planner, both uncertainty-driven rules reduced the steps to the quota relative to uniform priorities, by 12.6% on the corridor and 21.5% on the branching task, and lowered the mean width at 80000 steps by 0.087 and 0.147 (Table 7). At first sight this looked like a success for geometry, but it was not. Per-row accounting for one seed traced the saving to a few rows whose execution could be postponed by looping on known rows, which points to the planner rather than to the geometry. With equal terminal payoffs, the stopped recursion (45) is indifferent between executing an unresolved row now and doing so after a loop on a known row, as long as the loop fits in the horizon. Breaking that tie toward the smallest action index delays the execution to the end of the horizon. Source-based priorities alternate between rows with similar counts and so rarely meet the tie.

To isolate the effect, we reran all three rules with ties broken toward executing an unresolved row. This diagnostic was designed after the specified results had been analyzed, and we

TABLE 6: V6: mean supremum width of the baseline certificate at four sample counts per row and, at $n = 131072$, of the diameter-refined ($\bar{D}$) and support-refined certificates (V8) and of the simultaneous bootstrap band, which was computed for $S \leq 32$; mean number of anchors $|C|$ at $r = 0.2$, largest ℓ_∞ distortion κ, and median wall-clock time of one baseline certificate ($d^{\pm}$ and U_c), all at $n = 131072$. 100 seeds per size; every certified interval covered the true metric at every pair and checkpoint.

	Baseline width at n				Width at $n = 131072$					
S	2048	8192	32768	131072	$\bar{D}$	Support	Band	$\lvert C\rvert$	κ	Time (s)
16	0.687	0.377	0.198	0.103	0.066	0.059	0.006	4.00	0.060	0.06
32	0.700	0.385	0.202	0.105	0.068	0.060	0.007	4.00	0.062	0.14
64	0.710	0.391	0.204	0.107	0.069	0.061	–	4.00	0.063	0.28
128	0.718	0.396	0.207	0.108	0.070	0.061	–	4.00	0.063	1.39

TABLE 7: V7: interaction steps to a quota of 4096 samples per row, with 95% seed-bootstrap intervals, and mean pairwise certified width after 20000, 40000, and 80000 steps; 32 matched seeds per row of the table, all completed. "Local" denotes source-only priorities. "First action" is the specified planner, which breaks ties toward the smallest action index; "Unresolved row" is the post hoc diagnostic that breaks ties toward executing an unresolved row.

				Mean width after		
Task	Priorities	Ties	Steps to quota [95% interval]	20000	40000	80000
Corridor	Uniform	First action	227,566 [227,493, 227,648]	1.000	1.000	0.926
Corridor	Local	First action	198,871 [198,795, 198,955]	0.981	0.931	0.839
Corridor	Propagated	First action	198,887 [198,815, 198,967]	0.981	0.931	0.839
Corridor	Uniform	Unresolved row	194,798 [194,725, 194,880]	0.981	0.972	0.843
Corridor	Local	Unresolved row	194,775 [194,699, 194,859]	0.979	0.927	0.837
Corridor	Propagated	Unresolved row	194,791 [194,719, 194,871]	0.979	0.927	0.837
Branching	Uniform	First action	203,598 [203,480, 203,721]	0.996	0.982	0.960
Branching	Local	First action	159,839 [159,779, 159,899]	0.989	0.945	0.813
Branching	Propagated	First action	159,840 [159,779, 159,902]	0.989	0.945	0.812
Branching	Uniform	Unresolved row	159,894 [159,833, 159,955]	0.988	0.961	0.894
Branching	Local	Unresolved row	159,841 [159,779, 159,903]	0.989	0.945	0.813
Branching	Propagated	Unresolved row	159,840 [159,777, 159,903]	0.989	0.945	0.810

report it as post hoc; it leaves Theorem 17 untouched, because the analysis admits any maximizing policy. Uniform priorities then needed 14.4% and 21.5% fewer steps than before, and the three rules completed within 0.01% and 0.03% of each other. The anytime certificate still differed. By balancing counts across rows instead of completing rows one at a time, source-based priorities lowered the mean width by 0.045 after 40000 steps on the corridor and by 0.081 after 80000 steps on the branching task, with paired 95% intervals excluding zero. Propagated and source-only priorities differed by at most 16 steps to the quota and by less than 0.01 in mean width at every budget. The picture is therefore clear: the gain in the anytime certificate came from the local source, and the larger effect on total cost came from tie-breaking.

H. V8: SHARPER CERTIFICATES AND CERTIFIED DECISIONS

V8 recomputes the certificates of V1 on the frozen samples, and those of V6 on regenerated, identical samples, with the refinements of Proposition 12. The diameter-refined certificate uses KL reward intervals and iterates $\bar{D} = \|d^+\|_\infty$; the support-refined certificate also uses the known successor labels of each row, which our ring and lumpable models share with RiverSwim. Neither spends additional failure probability. Their widths appear in Tables 3 and 6, and on RiverSwim in Table 9.

Every refined interval covered the true metric at every pair and checkpoint, in all 500 V1 runs of each model and in all 1200 V6 certificate runs. The gains are substantial. On the high-discount ring at 131072 samples per row, the support-refined certificate is 3.4 times narrower than the baseline (0.116 against 0.397), and diameter refinement alone gives a factor of 1.9; on the low-discount ring the factors are 1.8 and 1.5. Relative to the bootstrap band of V5, the gap shrinks from 131 to 38 at $\gamma = 0.9$ and from 24 to 13 at $\gamma = 0.6$.

Even wide intervals can settle the questions that state aggregation actually asks. In the lumpable models of V6, every true distance is either zero or at least 0.259, so with $\varepsilon = 0.15$ a pair can be certified mergeable ($d^+ \leq \varepsilon$) or certified distinct ($d^- > 0$). Table 8 reports, for the baseline and support-refined certificates and for a simultaneous bootstrap band built as in V5, the percentage of bisimilar pairs certified mergeable, the percentage of distinct pairs certified distinct, and the number of runs with at least one wrong decision.

The certificates never made a wrong decision, in any run, model, or checkpoint, exactly as Theorem 5 guarantees on

TABLE 8: V8: certified merge and split decisions on the lumpable models of V6 with $\varepsilon = 0.15$, 100 seeds. "Merge" is the percentage of bisimilar pairs with $d^+ \le \varepsilon$, "split" the percentage of distinct pairs with $d^- > 0$, and "wrong" the number of runs in which some bisimilar pair was declared distinct or some distinct pair mergeable. The bootstrap band uses the same rules with its own endpoints.

		Baseline			Support-refined			Bootstrap band		
S	n	Merge	Split	Wrong	Merge	Split	Wrong	Merge	Split	Wrong
16	8,192	8	100	0	100	100	0	100	100	63
16	32,768	100	100	0	100	100	0	100	100	66
16	131,072	100	100	0	100	100	0	100	100	64
32	8,192	2	100	0	100	100	0	100	100	87
32	32,768	100	100	0	100	100	0	100	100	87
32	131,072	100	100	0	100	100	0	100	100	84

its event. Every distinct pair was certified distinct from 2048 samples per row on. The support-refined certificate certified 100% of the bisimilar pairs as mergeable at 8192 samples per row for $S = 16$ and 97% for $S = 128$, where the baseline managed 8% and 0%. The bootstrap band behaved very differently. It declared some bisimilar pair distinct in 63 to 68 of 100 runs for $S = 16$ and in 84 to 87 for $S = 32$, at every sample count, and it covered the true metric at every checkpoint in only 2 and 0 of 100 runs. More data does not cure this. The plug-in distance at a bisimilar pair is biased upward, because it maximizes noisy differences over actions, and both this bias and the half-width of the band scale as $n^{-1/2}$. A heuristic interval is thus least reliable precisely at the pairs that aggregation must merge, whereas the certificate is reliable there by construction. We regard this as the most practically important finding of the evaluation.

I. V9: RIVERSWIM

The models so far were designed by us, and a skeptical reader may ask whether the findings survive on a standard benchmark. V9 therefore moves to RiverSwim, the hard-exploration chain introduced with MBIE [14], in the six-state variant of [41] with $\gamma = 0.9$. Some rows have three successors, so every pairwise transport problem is solved exactly over the 81 spanning-tree bases of the 3×3 transportation polytope. The true metric has diameter 0.384.

All three certificates covered the true metric in every run and at every checkpoint (Table 9). Without support knowledge the intervals stay wide at this discount, but knowing the transition graph changes the picture considerably. At 131072 samples per row the support-refined certificate is 7.0 times narrower than the baseline, and it certifies 100% of the pairs as distinct, against 67% for the baseline. The bootstrap band remained about 15 times narrower than the sharpest certificate, but it covered all five checkpoints in only 62 of 100 runs.

RiverSwim earned its place as a benchmark because undirected exploration rarely reaches its far end. With resets to label 0, horizon $H = 20$, and a quota of 256 retained samples per row, optimistic navigation completed 32 of 32 runs, in 25,469 interaction steps on average. Random navigation also completed 32 of 32 runs within the cap of 10^6 steps, but it needed 867,613 steps on average, 34 times as many. The statistical quota is the same for both controllers; what makes it affordable on this benchmark is the learned navigation of Theorem 17.

TABLE 9: V9: RiverSwim with $\gamma = 0.9$, 100 seeds. Mean supremum width of the baseline, diameter-refined ($\bar{D}$), and support-refined certificates, all of which covered the true metric at every pair in every run, and coverage and width of the simultaneous bootstrap band (nominal level 0.9). The baseline and $\bar{D}$ certificates assume only $\Gamma = 3$; the support-refined one also uses the known successor labels of each row.

	Certified width			Bootstrap band	
n	Base	$\bar{D}$	Support	Cov.	Width
512	1.000	1.000	1.000	0.930	0.047
2,048	1.000	1.000	0.469	0.870	0.023
8,192	0.867	0.686	0.188	0.920	0.012
32,768	0.547	0.264	0.086	0.910	0.006
131,072	0.292	0.120	0.042	0.900	0.003

J. V10: ADJOINT PRIORITIES AND A TARGETED QUERY

V10 tests the adjoint priorities of Section V in the two uses that it distinguishes. For the full certificate, four rules (uniform, source-only, forward-propagated, and adjoint with uniform weight) run on the two V3 tasks and on RiverSwim with $\gamma = 0.6$, under the V7 protocol with ties broken toward unresolved rows, on 32 matched seeds (Table 10).

All rules reached the quota with almost the same interaction, within 0.13%, 0.04%, and 0.01% of each other on the three tasks, so once again priorities acted only on the anytime certificate. There, uncertainty-driven priorities helped on every task, and most on RiverSwim, where source-only priorities lowered the mean width at 320000 steps by 0.184 relative to uniform priorities. Geometry, however, added no consistent gain on top of the local source. Adjoint priorities lowered the final mean width on the branching task by 0.065 relative to source-only priorities, with paired interval $[-0.079, -0.057]$, but raised it by 0.025 at 20000 steps on the corridor, interval $[0.021, 0.028]$, and changed it by less than 0.01 on RiverSwim.

TABLE 10: V10, full certificate: interaction steps to a quota of 4096 samples per row and mean pairwise certified width at the last three budgets (20000, 40000, and 80000 steps on the V3 tasks; 80000, 160000, and 320000 on RiverSwim with $\gamma = 0.6$). 32 matched seeds per row, all completed; ties are broken toward unresolved rows for every rule.

			Mean width at budget		
Task	Rule	Steps	3rd	4th	5th
Corridor	Uniform	194,824	0.981	0.972	0.843
Corridor	Source	195,064	0.953	0.929	0.831
Corridor	Propagated	194,801	0.953	0.929	0.831
Corridor	Adjoint	194,872	0.978	0.928	0.836
Branching	Uniform	159,839	0.987	0.961	0.891
Branching	Source	159,771	0.970	0.959	0.866
Branching	Propagated	159,770	0.970	0.959	0.866
Branching	Adjoint	159,794	0.973	0.954	0.800
RiverSwim	Uniform	406,743	0.870	0.739	0.561
RiverSwim	Source	406,686	0.870	0.625	0.377
RiverSwim	Propagated	406,686	0.870	0.625	0.371
RiverSwim	Adjoint	406,706	0.870	0.625	0.372

TABLE 11: V10, targeted query: interaction steps until labels 4 and 5 are certified distinct ($d^- > 0$), with 95% seed-bootstrap intervals, and the share of retained samples taken at the two target labels. The query was answered in all 32 matched seeds of every row.

Task	Rule	Steps [95% interval]	Target %
RiverSwim	Count	3,293 [3,252, 3,337]	7
RiverSwim	Adjoint	3,285 [3,243, 3,322]	7
Corridor	Count	4,809 [4,445, 5,192]	16
Corridor	Adjoint	3,641 [3,425, 3,877]	20

Forward-propagated priorities never differed from source-only ones by more than 0.006 in mean width.

For a targeted query, we ask whether labels 4 and 5 are behaviorally distinct, on the corridor and on RiverSwim with $\gamma = 0.6$, and stop when the baseline lower endpoint of that pair exceeds zero. A geometry-free count rule, which prefers rows with large radii, is compared with adjoint priorities weighted on the target pair; both use the same planner and the same retained data (Appendix E).

On the corridor, adjoint priorities answered the query in 24% fewer steps than the count rule (Table 11). The paired difference was 1168 steps, with a 95% interval of $[-1340, -1007]$, and a larger share of the samples went to the target labels. On RiverSwim the two rules were indistinguishable, with a paired difference of -8 steps and interval $[-54, 31]$. The reason is not hard to see. On RiverSwim, reaching labels 4 and 5 consumes most of every episode, and both rules sample the target rows whenever the agent arrives. Geometry pays off for a targeted query when rows that do not influence it are cheap to sample and would otherwise absorb the budget; it cannot shorten the walk to rows that must be reached in any case.

K. V11: THE NEURAL INSTANTIATION ON A BLOCK MDP

V11 evaluates the representation and width components of Appendix F in a setting where the true metric is known. The latent dynamics are the low-discount ring of V1 ($S = 6$, $\gamma = 0.6$). Each 24-dimensional observation adds a five-state exogenous random walk, embedded at 1.5 times the scale of the latent signal, and Gaussian noise with standard deviation 0.1; by Theorem 22, the true metric ignores the distractor. A generative simulator returns $m = 8$ successor observations and rewards per query, and a bootstrap ensemble of five resamples forms the targets of Appendix F, with exact transport between successor sets under the frozen encoder. We compare the interval pipeline (width critic and the interval-weighted loss (84)), the same pipeline with point targets and zero width, which is a sampled analogue of deep bisimulation [4], and raw observation distances rescaled by the least-squares factor fitted to the true metric, a reference that favors the raw baseline. Each learned method uses five seeds and 3000 updates, with hyperparameters fixed before the run and not tuned.

Both learned encoders suppress the distractor relative to raw observations (Table 12). The ratio of mean distances between and within latent states rises from 1.30 to 2.10 with point targets and to 1.83 with interval targets, and the mean distortion falls from 0.110 to 0.093 and 0.102. Neither encoder reaches zero at equal latent states, however: observations of the same latent state stay 0.208 and 0.240 apart on average, against a true distance of zero. A post hoc diagnostic locates most of this floor in the targets rather than in the encoders. With the true metric as ground cost and $m = 8$, the sampled point target averages 0.132 at bisimilar pairs, and its bias falls to 0.067 and 0.034 at $m = 32$ and $m = 128$, consistent with an $m^{-1/2}$ rate. The mechanism is the same one we met in V8. A maximum over actions of noisy differences is positive even when every difference has mean zero, so plug-in distances are biased upward at bisimilar pairs, and any method that regresses onto sampled bisimulation targets inherits this bias. The interval pipeline did not improve distortion over point targets. Its surrogate intervals contained the true distance for 0.974 of held-out pairs, but their mean width of 0.606 exceeds the diameter of the true metric, so this coverage carries little information. In short, the evaluation supports the encoder side of the extension and quantifies a bias that all sampled targets share, but it also shows that the width critic, as specified, does not yet deliver informative uncertainty.

IX. DISCUSSION AND LIMITATIONS

The confidence interval separates the behavioral geometry from our uncertainty about it, and each of its parts has a distinct use. The midpoint is a regression target, the upper endpoint supports a cover and an anchor map, and the width records both unresolved local rows and unresolved coupled futures. The bounded recursive majorant localizes this width without weakening the certificate, which continues to rest on contraction alone. Three questions remain, and we take them in turn.

TABLE 12: V11: held-out results on the block MDP, means over five seeds after 3000 updates; 2000 random pairs of 300 held-out observations per seed. Distortion is $|\,\|f(o) - f(o')\| - d_{\sim}|$; "same" and "different" are mean embedding distances between observations with equal and with distinct latent states, whose true values are 0 and 0.376 on average. Coverage is the fraction of held-out pairs whose surrogate interval $[L_\psi, H_\psi]$ contains the true distance; it is not a certificate.

Method	Mean dist.	Max dist.	Same	Different	Coverage	Width
Raw observations (oracle scale)	0.110	0.364	0.246	0.321	–	–
Point targets	0.093	0.337	0.208	0.436	–	–
Interval targets (App. F)	0.102	0.411	0.240	0.440	0.974	0.606

When are the intervals useful in practice? We see three settings. First, a certificate settles questions long before its width is small. In V8 and V9 the lower endpoint certified most distinct pairs as distinct, and in V8 the upper endpoint certified bisimilar labels as mergeable, without a wrong decision in any run, whereas a bootstrap band built from the same samples wrongly separated bisimilar labels. Abstraction and state aggregation consist of exactly such decisions. Second, when the support of each row is known, as it is in RiverSwim, in gridworlds, and in many simulators, the refinement of Proposition 12 narrows the intervals by up to 7.0 times at no cost in confidence. Third, generative simulators make 10^4 to 10^5 samples per row inexpensive, and in that regime the sharpest certified widths of V8 and V9 fall to between 0.04 and 0.12. Where samples are scarce and the discount is high, the intervals remain wide. In that regime a bootstrap band is the more informative heuristic, provided one can live without a guarantee.

Where does the remaining conservatism come from? Even after the refinement, the certificates are 13 to 38 times wider than the bootstrap band on the ring models and about 15 times wider on RiverSwim. Most of this gap is the price of the guarantee itself. The union over rows and visit counts places $\ell_n \approx 30$ under every radius at $n = 131072$, the intervals must hold at every snapshot rather than at one, and the recursion bounds the worst coupled future rather than a typical one. The band, for its part, covered every checkpoint in only 58 to 62% of runs on the ring models and RiverSwim, and in at most 2% on the lumpable models, where bisimilar pairs exist. The rest of the gap is due to rectangular confidence sets, which ignore the fact that a single model must explain all pairs at once. Confidence sets shared through a common model, and couplings optimized jointly across pairs, could narrow it, provided their validity under adaptive data can be proved.

What does geometry contribute to acquisition? The interaction theorem holds for any priorities in $[1/2, 1]$, so this is an empirical question, and the answer is more modest than we initially expected. Acquisition cost depended strongly on navigation, by a factor of 34 on RiverSwim, and on how the planner breaks ties. Source-based priorities improved the anytime certificate by balancing counts across rows, by up to a third on RiverSwim, but neither forward-propagated nor adjoint priorities added a consistent gain on top. For a targeted query, influence on the queried pair saved 24% of the interaction on the corridor, where cheap irrelevant rows would otherwise absorb the budget, and nothing on RiverSwim, where the walk to the queried labels dominates. Geometry thus helps acquisition when it can tell the agent which cheap rows to stop sampling; it does not shorten the walk to rows that must be reached anyway.

Four structural limitations remain, and the lower bounds and counterexamples above make each of them precise. The dimension of the true metric does not bound the number of rows that can be perturbed independently. Equivalence under extrinsic rewards does not identify arbitrary goals. A general pseudometric need not embed exactly in a low-dimensional Euclidean space. And local sampling cannot resolve futures that leave the sampled region. Beyond these, the two-clock construction is sharp on its family, but we do not prove minimax optimality for the navigation factors of our upper bound. Continuous successor laws bring their own transport-estimation complexity, and an optimistic learned metric does not remove the observation dimension from the concentration rates. Finally, the neural instantiation transfers the encoder side of the construction but not yet its uncertainty (V11). Debiased targets, calibrated width critics, and image-based tasks with learned policies are the natural next steps for that extension.

X. CONCLUSION

Treating the bisimulation metric as an uncertain object yields computable confidence intervals for finite MDPs, a pair-space majorant that tracks where their width comes from, and certified anchor representations. Under reset reachability, an agent can obtain the model and the navigation policies that these outputs require from interaction alone. Two lower bounds mark the limits of the approach: geometric compression is not model identification, and on a controlled family, certifying a cover costs two more powers of the accuracy than finding one.

Eleven pre-specified studies support the theory and measure its price. The certificates covered the true metric in every recorded run, from four labels to 128 and on RiverSwim. A diameter refinement on the same confidence event narrowed them by up to 7.0 times, and they certified merge and split decisions without error where bootstrap intervals erred. Navigation and planner tie-breaking governed the cost of acquisition, while influence priorities helped a targeted query only where irrelevant rows were cheap. The neural

instantiation learned a partly invariant encoder but not yet informative uncertainty, and closing that gap is the main open step toward rich observations.

CODE AND DATA AVAILABILITY

The code, frozen experiment specifications, seed-level records, and run manifests that reproduce every result in this paper are publicly available at https://github.com/Adityasikarwar3107/epistemic-bisimulation-geometry [8]. A verification script checks every stored record against the hashes in its run manifest.

APPENDIX A COMPLETE PROBABILITY AND OPERATOR ARGUMENTS

A. ANYTIME CONFIDENCE PROOF

Fix a state-action row and imagine an infinite stream of independent samples from its stationary reward-successor law. Each time the agent visits the row, it consumes the next sample of the stream. This construction has the same law as adaptive interaction with the MDP, so confidence that holds simultaneously for every prefix of every stream also holds at the visit counts the agent happens to reach.

For $n \geq 1$, Hoeffding's inequality gives

$$\mathbb{P}\Big(|\hat{r}_n - r| > R_{\max}\sqrt{\ell_n/(2n)}\Big) \leq 2e^{-\ell_n}. \tag{62}$$

For a successor law supported on at most Γ labels, the multinomial deviation inequality gives

$$\mathbb{P}\big(\|\hat{P}_n - P\|_1 > u\big) \leq 2^{\Gamma} e^{-nu^2/2}. \tag{63}$$

With $u = 2\sqrt{(\Gamma \log 2 + \ell_n)/(2n)}$ the right-hand side is at most $e^{-\ell_n}$. When a radius is clipped at the largest possible error, the corresponding failure event is empty, and the convention at $n = 0$ is deterministic for the same reason.

There are SA rows. Summing both tails over all rows and all $n \geq 1$,

$$\begin{aligned}\mathbb{P}(\mathcal{E}^c) &\leq SA \sum_{n \geq 1} 3 \cdot \frac{3\delta}{4SA\pi^2 n^2} \\ &= \frac{3\delta}{8} \leq \delta,\end{aligned} \tag{64}$$

using $\sum_{n \geq 1} n^{-2} = \pi^2/6$. The constants leave deliberate slack. The number of observed successors cannot replace the fixed bound Γ, because a successor not yet observed can have positive mass.

APPENDIX B TRANSPORT AND METRIC PRESERVATION

A. PRIMAL TRANSPORT STABILITY PROOF

For any feasible coupling Λ, $|\langle \Lambda, h - \tilde{h} \rangle| \leq \left\| h - \tilde{h} \right\|_{\infty}$, since Λ has total mass one. Evaluating $\tilde{h}$ at an h-optimal plan gives one inequality, and exchanging the two costs gives the other. The same evaluation argument proves monotonicity.

For the perturbation of marginals, let (X, Y) be distributed according to an optimal coupling of (μ, ν). Conditionally on (X, Y), couple X maximally with $\tilde{X} \sim \tilde{\mu}$ and Y maximally with $\tilde{Y} \sim \tilde{\nu}$. This gives a feasible coupling of $(\tilde{\mu}, \tilde{\nu})$ with

$$\mathbb{P}((X, Y) \neq (\tilde{X}, \tilde{Y})) \leq \mathrm{TV}(\mu, \tilde{\mu}) + \mathrm{TV}(\nu, \tilde{\nu}). \tag{65}$$

Both costs take values in $[0, R_{\max}]$, so their expectations under the two couplings differ by at most $R_{\max}$ times this probability. The perturbed optimum is therefore at most the original optimum plus the stated error, and reversing the roles gives the absolute bound. The argument never uses a triangle inequality for h.

B. FULL EPISTEMIC SANDWICH PROOF

Each operator returns a symmetric, nonnegative cost with zero diagonal, bounded by $R_{\max}$. For the lower operator this holds because each positive part is at most the corresponding uncorrected reward or transport term, and the convex combination of those terms is at most $R_{\max}$. Both operators therefore map $\mathfrak{C}$, a closed subset of a finite-dimensional Banach space, into itself. For every pair,

$$|\mathcal{T}^{\pm}[h](s, s') - \mathcal{T}^{\pm}[\tilde{h}](s, s')| \leq \gamma \left\| h - \tilde{h} \right\|_{\infty}, \tag{66}$$

so Banach's theorem gives a unique fixed point for each operator.

On $\mathcal{E}$, the triangle inequality for the reward estimates and Lemma 4 give the operator comparison. The true operator takes values in $[0, R_{\max}]$, so clipping the upper operator cannot reverse it. Iterating the upper operator from $d_{\sim}$ gives $d_{\sim} \leq \mathcal{T}^{+}[d_{\sim}] \leq (\mathcal{T}^{+})^2[d_{\sim}] \leq \cdots$, with limit d^{+}; iterating the lower operator from $d_{\sim}$ gives a decreasing sequence with limit d^{-}. This proves the sandwich.

It remains to show that d^{+} satisfies the triangle inequality. For a pseudometric h, transport between probability measures is a pseudometric: glue optimal couplings of (μ, ν) and (ν, ξ) into a coupling of three variables and use $h(X, Z) \leq h(X, Y) + h(Y, Z)$. For each action define

$$c_a(s) = (1 - \gamma)\beta^r(s, a) + \gamma R_{\max}\beta^P(s, a), \tag{67}$$

so that the confidence term added off the diagonal is $c_a(s) + c_a(s')$. For three distinct labels, this star cost satisfies the triangle inequality because the two-step sum contains the extra nonnegative term $2c_a(x)$; cases with coinciding labels are immediate. The reward distance, the transport distance, and the star cost are pseudometrics, and they enter with total weight one. A maximum over actions of pseudometrics is a pseudometric. Finally, if g is a pseudometric then so is $\min\{R_{\max}, g\}$: either the two truncated terms on the right sum to at least $R_{\max}$, or the truncation does not affect the ordinary triangle bound. Iterating $\mathcal{T}^{+}$ from the zero pseudometric and passing to the limit shows that d^{+} is a pseudometric.

C. VALUE BOUND AND ITS POLICY LIMITATION

Let $v^* = (1 - \gamma)V^*$, whose Bellman operator is $Bv(s) = \max_a[(1 - \gamma)r(s, a) + \gamma \mathbb{E}_{P_{s,a}} v]$. If v is 1-Lipschitz in $d_{\sim}$, then every feasible coupling gives $|\mathbb{E}_{\mu} v - \mathbb{E}_{\nu} v| \leq \langle \Lambda, d_{\sim} \rangle$. Minimizing over couplings and comparing the maxima over

actions yields $|Bv(s) - Bv(s')| \le \mathcal{T}[d_\sim](s,s') = d_\sim(s,s')$. Starting value iteration at zero and passing to the limit proves the bound. The proof for a fixed policy is identical on the induced one-action model.

The bound fails for arbitrary policies with $d_\sim$. Take two labels with identical reward vectors $(0,1)$ that self-loop under both actions, so that $d_\sim = 0$. A policy that takes the zero-reward action at one label and the unit-reward action at the other has value difference $1/(1-\gamma)$. The policy's own induced metric does reflect this difference.

D. PAIR MAJORANT AND OCCUPANCY FORMULA

For each action and pair, the lower fixed point is at least $q_a(d^-) - b_a$, even when this expression is negative, and the upper fixed point is at most $\max_a[q_a(d^+) + b_a]$. The elementary inequality for differences of maxima, together with (23), proves the pair Bellman inequality. The map $\mathcal{F}$ is monotone and γ-contractive on bounded functions of pairs, and its rewards are bounded and nonnegative, so iterating it from Δ gives an increasing sequence that converges to U.

For a fixed stationary pair policy, summing the geometric series $\sum_{j\ge 0} \gamma^j (P_\Lambda^\sigma)^j$ gives (26). A finite discounted MDP has a stationary deterministic optimal policy whose value is the unique Bellman fixed point, which proves (25). Diagonal plans stay on diagonal pairs and collect zero reward, so $U(s,s) = 0$.

E. REFINED ENDPOINTS

Monotonicity and contraction follow as in Appendix B-B, because the constants π_a and $\bar{D}$ do not depend on h. On $\mathcal{E}$, the true reward difference lies in $[\underline{g}_a, \bar{g}_a]$. For the transport term, let $h = d_\sim$, a pseudometric. Take an optimal coupling of $(\hat{P}_{s,a}, \nu)$ and a maximal coupling $(\hat{X}, X)$ of $\hat{P}_{s,a}$ and $P_{s,a}$. Gluing them gives a coupling of $(P_{s,a}, \nu)$, and the triangle inequality $h(X,Y) \le h(\hat{X},Y) + h(X,\hat{X})$ yields $\mathrm{OT}_h(P_{s,a},\nu) \le \mathrm{OT}_h(\hat{P}_{s,a},\nu) + D_{s,a}\,\mathrm{TV}(P_{s,a},\hat{P}_{s,a})$, because X and $\hat{X}$ both lie in $\Sigma_{s,a}$. The reverse inequality and the same step for the second marginal give $|\mathrm{OT}_{d_\sim}(P_{s,a},P_{s',a}) - \mathrm{OT}_{d_\sim}(\hat{P}_{s,a},\hat{P}_{s',a})| \le \pi_a/\gamma$. Hence $\mathcal{T}_D^-[d_\sim] \le \mathcal{T}[d_\sim] = d_\sim \le \mathcal{T}_D^+[d_\sim]$, where the clip at $\bar{D}$ is harmless because $d_\sim \le \bar{D}$. Iterating the monotone contractions from $d_\sim$ proves the sandwich. For the pseudometric property, $\bar{g}_a$ with zero diagonal satisfies the triangle inequality, since $\bar{r}_s - \underline{r}_{s''} \le (\bar{r}_s - \underline{r}_{s'}) + (\bar{r}_{s'} - \underline{r}_{s''})$, and π_a is a star cost as in Appendix B-B; the remaining steps of that argument apply unchanged, so d_D^+ is a pseudometric. Under the stated inclusions both refined operators lie between the original ones pointwise, which orders the fixed points. Finally, the proof of Theorem 6 uses only $d^+ \le \max_a\{(1-\gamma)\bar{g}_a + \gamma\hat{w}_a(d^+) + \pi_a\}$ and $d^- \ge \max_a\{(1-\gamma)\underline{g}_a + \gamma\hat{w}_a(d^-) - \pi_a\}$, which hold here, so the same argument gives the stated source.

APPENDIX C COMPLETE CLOSED-LOOP PROOF

A. STOPPED SIMULATION BOUND

Fix a policy and couple each true known-row transition maximally with its empirical counterpart. Until the first mismatch, the two systems share the same state, action, remaining horizon, and stopping rule. Frozen terminal priorities at unresolved rows lie in $[1/2, 1]$. A union bound over at most H transitions shows that the expected terminal payoffs of the two systems differ by at most $H\alpha_*$, uniformly over policies, because the row error is uniform. For any unresolved row, a true policy reaches it with probability at least p or stops earlier at another unresolved row, so its expected payoff is at least $p/2$. The empirical optimum is therefore at least $p/2 - H\alpha_*$, and the executed policy has true expected payoff at least $p/2 - 2H\alpha_* \ge p/4$. Its probability of sampling an unresolved row is at least this payoff.

B. AN ADAPTED SUCCESS-COUNT BOUND

Let $Q = SAn_0$ be the number of retained successes required and let $q = p/4$. For exploratory episode k, let $X_k = 1$ if an unresolved row is executed and $X_k = 0$ otherwise. To avoid conditioning on a future confidence event, we work with an auxiliary process that sets $X_k = 1$ after completion or after the first confidence failure. On every earlier good history, its conditional success probability is at least q, so for $\lambda = \log 2$,

$$\begin{aligned}\mathbb{E}[e^{-\lambda X_k} \mid \mathcal{H}_{k-1}] &\le 1 - q + qe^{-\lambda} \\ &= 1 - q/2 \le e^{-q/2}. \end{aligned} \tag{68}$$

Iterating conditional expectations and applying Markov's inequality,

$$\mathbb{P}\left(\sum_{k=1}^{N} X_k < Q\right) \le \exp\{(\log 2)Q - pN/8\}. \tag{69}$$

With $N = \lceil 8(Q + \log(2/\delta))/p \rceil$ the right-hand side is at most $\delta/2$. On the model confidence event the auxiliary and actual processes agree until termination, and adding the failure probability $\delta_m = \delta/2$ of that event proves the bounds on episodes and interaction steps. If each reset costs at most c_{reset} physical steps, H is replaced by $H + c_{\mathrm{reset}}$ in the total.

C. RESOLUTION, COMPRESSION, AND GOAL POLICIES

At termination every row holds exactly n_0 retained samples. By (42), $\beta^r \le R_{\max}\alpha_*$ and $\beta^P \le \alpha_*$, and (10) gives $b_a \le 2R_{\max}\alpha_*$. Hence

$$\left\|d^+ - d^-\right\|_\infty \le \frac{4R_{\max}\alpha_*}{1-\gamma} \le \varepsilon/8. \tag{70}$$

Each computed endpoint is padded by a residual-plus-transport margin of at most $\varepsilon/16$, so the total widening is at most $\varepsilon/8$ and the reported width is at most $\varepsilon/4$. Path closure of the padded upper endpoint is still an upper bound and can only reduce the width; we call the resulting endpoints L and U. Applying Theorem 13 to L, U with $r = \varepsilon/2$, the separation scale is at least $\varepsilon/4$, so the number of centers is at most $\mathcal{N}_\sim(\varepsilon/8)$ and the distortion is at most ε. Every center has been visited, because every row has been sampled.

For a center g, plan the empirical finite-horizon hitting policy with terminal condition $V_0(s) = \mathbf{1}\{s = g\}$, treating g as absorbing for evaluation. The coupling argument over $H-1$

transitions gives a true success probability of at least $p-2(H-1)\alpha_* \geq p/2$. This policy is learned; no goal reward is inferred from the extrinsic bisimulation distance. The statement about Euclidean fitting follows from Proposition 14 applied to the reported interval.

D. EXACT QUOTIENT COMPOSITION

For an exact homomorphism, the distribution of decoded successors and the mean reward are constant on each block, under every action. Decoded successor samples collected at different raw labels of one block therefore share a conditional transition law. The homomorphism fixes the reward means but not necessarily the full reward distributions; bounded rewards minus their common decoded mean form martingale differences, so the conditional Hoeffding bound replaces the independent-stream argument for pooled observations. The concentration of decoded successors and the union bound over visit counts are unchanged, and no common reward-noise distribution within a block is assumed.

Every raw transport coupling projects to a decoded coupling. Conversely, a decoded coupling lifts to a raw one by conditioning the raw successor distributions on each block, with the same cost for a block-constant ground metric. The lifted quotient metric is therefore the unique raw fixed point. The finite-state proof applies to the decoded rows and pulls back to raw states. A decoder learned in an earlier phase contributes its own confidence and sample cost, and a union bound proves Corollary 18.

APPENDIX D COMPLETE LOWER-BOUND CONSTRUCTIONS

A. GEOMETRY-ONLY IDENTIFICATION OBSTRUCTION

With absorbing self-loops, the transport cost for a pair equals its own ground cost, so a fixed point satisfies $d(i,j) = (1-\gamma)\max_a |r(i,a) - r(j,a)| + \gamma d(i,j)$, which proves (51). In the base instance all such distances are zero.

Let $N_{i,a}$ be the number of generative queries to row (i,a) before stopping, and let B_i be the event that the reported upper endpoint for $(i, \text{reference})$ is below 2ε. Under the base instance, valid intervals of width at most ε imply B_i, so $\mathbb{P}_0(B_i) \geq 1-\delta$. Under the alternative that changes (i,a), validity requires the upper endpoint to be at least 4ε, so $\mathbb{P}_{i,a}(B_i) \leq \delta$.

The chain rule for the relative entropy of an adaptively stopped experiment, followed by data processing onto B_i, gives

$$\begin{aligned}\mathbb{E}_0[N_{i,a}]\ \mathrm{kl}(1/2, 1/2+4\varepsilon) &\geq \mathrm{kl}(\mathbb{P}_0(B_i), \mathbb{P}_{i,a}(B_i)) \\ &\geq \mathrm{kl}(1-\delta, \delta), \end{aligned} \tag{71}$$

since all other row laws coincide between the two instances. For $\varepsilon \leq 1/16$,

$$\begin{aligned}\mathrm{kl}(1/2, 1/2+4\varepsilon) &= -\tfrac{1}{2}\log(1-64\varepsilon^2) \\ &\leq \frac{32\varepsilon^2}{1-64\varepsilon^2} \leq 64\varepsilon^2. \end{aligned} \tag{72}$$

Summing (71) over all SA nonreference rows gives the theorem. An infinite expected stopping time satisfies the bound trivially; otherwise the standard stopped change-of-measure inequality applies [34]. The base quotient has a single point, so its covering number is one and its doubling dimension is zero.

B. CONTROLLED TWO-CLOCK SEPARATION

By the self-loop formula, the metric is the largest difference of action means. Between grid points, every nonzero coordinate difference is at least 8ε, while the unknown action contributes a difference of at most 4ε; this proves the grid part of (55). Differences in the known coordinates between the reference and a grid point are below $1/2$, whereas $\mu_q > 1/2$, so the unknown action determines every distance to the reference.

Grid labels are at least 8ε apart, and the reference is more than 2ε from every grid point. An ε-cover built from visited labels must therefore contain every label, and a generative schedule visits each one in $M+1$ queries. Each lattice ball is covered by at most 5^d lattice balls of half the radius. A ball that contains both the reference and a grid point has radius at least $\min_q \mu_q > 0.7$, whose half exceeds the grid diameter $1/4$, so two centers cover it. The doubling dimension is therefore $O(d)$, uniformly in m.

For identification, compare the instance with all $\theta_q = -1$ to the alternative that flips a single sign. An interval of width ε for that reference distance cannot be valid for both values, which are 4ε apart. As above, each unknown row requires an expected number of samples of at least

$$\frac{\mathrm{kl}(1-\delta, \delta)}{\mathrm{kl}(3/4-2\varepsilon, 3/4+2\varepsilon)}. \tag{73}$$

The inequality $\mathrm{kl}(u,v) \leq (u-v)^2/[v(1-v)]$ and $\varepsilon \leq 1/64$ bound the denominator by $128\varepsilon^2$, and summing over the M rows proves the lower bound.

For the upper bound, draw

$$n = \left\lceil \frac{8}{\varepsilon^2} \log \frac{2M}{\delta} \right\rceil \tag{74}$$

Bernoulli samples from each unknown row. By Hoeffding's inequality and a union bound, $|\hat{\mu}_q - \mu_q| \leq \beta = \sqrt{\log(2M/\delta)/(2n)} \leq \varepsilon/4$ simultaneously for all q. Report $[\hat{\mu}_q - \beta, \hat{\mu}_q + \beta]$, clipped to $[0,1]$, for each reference distance. The known means are exact, and even over these intervals any two unknown means differ by at most $4\varepsilon + 4\beta \leq 5\varepsilon < 8\varepsilon$. Grid-to-grid distances are therefore known exactly, and every reference interval has width at most $\varepsilon/2$. The total sample count is Mn plus one reference visit, which proves the upper bound. Since $m = 1/(32\varepsilon)$, $M = \Theta(\varepsilon^{-d})$.

APPENDIX E EXPERIMENTAL AND NUMERICAL SPECIFICATIONS

A. SAMPLING DESIGN AND ACCESS MODELS

All configurations were fixed before execution, except the two post hoc diagnostics and the pilot-based designs named in Section VIII.

V1 uses 500 seeds per model, indexed 10000–10499. Independent, model-specific reward and successor streams generate nested binomial increments, which are exact sufficient statistics of the declared Bernoulli row laws. The count n is per state-action row, so the generative budget at count n is SAn samples, with no navigation. The two ring models are defined in (60) and Section VIII.

V2 uses seed indices 31000–31099. The dynamics and the current-row rewards are known, so their confidence radii are zero. Only the two future reward rows are sampled, which requires $2n$ Bernoulli observations at count n. Each seed uses the same future reward draws for all escape probabilities, and seeds are the independent units of aggregation. Local source, actual width, bounded pair majorant, global bound, and pair coverage are evaluated at every checkpoint.

V3 uses seed indices 51000–51031 on each task, with four matched priority and navigation variants. Common row streams give every variant the same first 64 reward and successor outcomes per row, and each planner sees only the observations it has acquired. Every allocated run contributes to the summary, with its completion status and interaction cost. The transitions of both tasks are given in Section VIII; for both, the Bernoulli rewards have means

$$r(i,0) = 0.05 + \frac{0.6i}{S-1},$$
$$r(i,1) = 0.1 + \frac{0.7(3i \bmod S)}{S-1}. \tag{75}$$

In the branching model, label $i < 3$ has child $2i+1+a$ under action $a \in \{0,1\}$. Both tasks reset to label 0. The random controller samples actions uniformly. Optimistic navigation follows (45) and, as fixed in the original specification, breaks ties toward the smallest action index; Section VIII-G shows why the tie-breaking rule adopted in Section V is preferable. Source-only priorities use the local-source score with the normalization of (44), and propagated priorities use the same normalization with U_c. Priorities start at one and are refreshed every 128 episodes. An empirical transition row with no observations is initialized to a self-loop and its mean reward to zero; the maximal radii at zero count cover this initialization. An episode stops at its first unresolved-row sample or at its horizon and then resets. Steps taken through known rows count as interactions even when their observations are not retained. Resets are free. Goal policies are evaluated within $H-1$ transitions, separately from the exploration horizon H.

V5 reuses the frozen V1 sufficient statistics at every checkpoint with $n \geq 128$, so all four interval methods see identical samples. The simulation-lemma interval is clipped to $[0,1]$ off the diagonal, with ρ capped at one; it is valid on the event of Lemma 2 because $\left\|\mathcal{T}[h] - \hat{\mathcal{T}}[h]\right\|_\infty \leq \|b\|_\infty$ for every $h \in \mathfrak{C}$, where $\hat{\mathcal{T}}$ is the operator of the empirical model. For each seed and checkpoint, 200 bootstrap replicates redraw every row's reward successes and first-successor count from binomial laws with the empirical means, which is the nonparametric bootstrap for these Bernoulli and two-point row laws [39]. The pointwise interval takes the 5th and 95th percentiles of the replicate metric at each pair. The simultaneous band takes the 90th percentile c of $\left\|\hat{d}^* - \hat{d}\right\|_\infty$ over replicates and reports $[\hat{d} - c, \hat{d} + c]$, clipped as above. Both have nominal level $0.9 = 1 - \delta$. Replicate seeds are derived from the V1 seed, the model, and the checkpoint.

V6 uses seed indices 80000–80099 for each $S \in \{16, 32, 64, 128\}$. Labels are split into four contiguous blocks of equal size. Block k has mean rewards $(0.1 + 0.25k,\ 0.85 - 0.25(3k \bmod 4))$; under action 0 it moves to block k with probability 0.7 and to block $k+1$ with probability 0.3, and under action 1 to block $k+2$ with probability 0.6 and to block $k+3$ with probability 0.4, indices modulo four. Each label draws its two successor labels uniformly from the designated blocks with a structure seed fixed by S, so every row has two successors and the model is exactly lumpable onto the four blocks. Rewards are Bernoulli. Nested prefixes of $2048, 8192, 32768, 131072$ samples per row are drawn as in V1. The anchor cover uses $r = 0.2$ at the final checkpoint, where every recorded width is below r. Certificate time is the median wall-clock time of five single-seed solves of $d^{\pm}$ and U_c at the final checkpoint, on one CPU process.

V7 uses seed indices 61000–61031 on the two V3 tasks with optimistic navigation and three priority rules: uniform ($v_t = 1$), source-only, and propagated. Every row must hold 4096 retained samples, priorities are refreshed every 512 episodes, and runs are capped at 10^6 steps. At the first episode boundary after 5000, 10000, 20000, 40000, and 80000 interaction steps, the snapshot endpoints are computed from all retained data with $\delta = 0.1$ and $\Gamma = 2$; a row with no samples has maximal radii. Common row streams are separate from those of V3 and are shared by the three rules. The quota and the budget grid were set after a pilot with seed 61000 on two task–rule pairs and a quota of 256, in which the recorded widths stayed at or near one; a timing check of the final configuration on the same seed preceded the full run. The pilot runs are not part of the reported summaries. The specified planner breaks ties toward the smallest action index. The post hoc diagnostic of Section VIII-G reruns the same seeds, tasks, and rules with ties broken toward executing an unresolved row, using a tolerance of 10^{-12}; its specification was written after the specified V7 results had been analyzed and is stored separately.

V8 regenerates the V6 statistics from the same seeds, which reproduces the V6 samples exactly. The KL intervals use the budget ℓ_n of (6); the diameter-refined certificate keeps Weissman radii with the declared Γ, whereas the support-refined one uses each row's own support size. Refinement stops when an iteration changes the padded upper endpoint by less than 10^{-9} or after 40 iterations. A pair counts as certified distinct when $d^- > 10^{-9}$. For $S \in \{16, 32\}$, a simultaneous bootstrap band is formed as in V5.

V9 uses the six-state RiverSwim variant of Osband et al. [41], introduced by Strehl and Littman [14]. The left action is deterministic. The right action stays at label 0 with probability 0.4 and otherwise advances; at interior labels it moves left,

stays, or advances with probabilities 0.05, 0.6, and 0.35; at the last label it moves left with probability 0.4 and otherwise stays. Bernoulli rewards have mean 0.005 for the left action at label 0 and one for the right action at label 5, and $\gamma = 0.9$. Generative runs use seeds 90000–90099 with multinomial successor counts, $\Gamma = 3$, and a bootstrap band built from multinomial and binomial resampling. The navigation runs use seeds 92100–92131.

V10 uses seeds 63100–63131 for the full certificate, with budgets of 20000 to 320000 steps and $H = 20$ on RiverSwim. The source-only score is the local source of the width recursion in Proposition 12, $(1-\gamma)(\bar{g}_a - \underline{g}_a) + 2\pi_a$ with baseline radii, which equals $2b_a$ of (10) unless the observed reward gap is smaller than e^r_a; V7 used $2b_a$, so the two studies are not compared directly. The targeted query uses seeds 64100–64131 and stops when the baseline lower endpoint of the target pair exceeds zero. Its planner may sample the current row with payoff v_t or, once the row holds 16 samples, cross it under the empirical law, and every executed transition is retained. The count rule scores a row by its own radius term $(1-\gamma)\beta^r + \gamma R_{\max}\beta^P$. The adjoint rule uses (48) with all weight on the target pair, retires a row with zero influence once it holds 16 samples, and breaks ties by a 10^{-3} multiple of the radius term. Both map scores to $[1/2, 1]$. Pilot runs on seeds 63000, 64000, 64001, and 92000 fixed these designs and are not reported.

V11 uses seeds 70000–70004 per method. The encoder and critic are two-layer SiLU networks with 64 hidden units and an eight-dimensional embedding, trained with Adam at learning rate 10^{-3} on 64 pairs per update, with target networks averaged at rate 0.01, $\beta_{cal} = 2$, and $\zeta = 0.05$. Successor sets of eight observations are matched by exact assignment, and each of five bootstrap heads resamples them with replacement. Evaluation uses 300 held-out observations per seed. The bias diagnostic draws 3000 random latent pairs per successor count with seed 71000.

B. SOLVERS AND ERROR ACCOUNTING

All arrays are float64. On rows with at most two successors, the feasible 2×2 coupling is parametrized by

$$z \in [\max\{0, p_0 + q_0 - 1\}, \min\{p_0, q_0\}]. \tag{76}$$

The transport objective is affine in z, so the minimizing endpoint is determined by the sign of $h_{00} - h_{01} - h_{10} + h_{11}$. Singleton supports are handled as cases with zero mass on one successor. Four coefficients apply the pair kernel without forming an AS^4 tensor. Independent checks compare this calculation with scalar primal transport and with linear programming on supported examples.

Fixed-point iteration stops when the step norm divided by $1-\gamma$ is at most 2×10^{-11}, after which the actual operator residual is evaluated and endpoint padding is set to that residual divided by $1-\gamma$. The possible suboptimality of the plan for the lower cost is bounded by twice the error of the lower fixed point, and the width majorant receives its own residual margin plus the endpoint padding. Coverage, midpoint, triangle, and majorant checks use the tolerance 2×10^{-8}. These are floating-point checks, not interval-arithmetic proofs that account for machine rounding.

At the final low-discount V1 checkpoint, the anchor audit builds the greedy cover of the upper metric with radius 0.25 and evaluates, for every pair, the infinity-norm representation distance, the covering radius, and the distortion. With all labels as anchors, it also checks that the representation distance equals the upper metric.

C. STATISTICAL SUMMARIES AND FAILED RUNS

For calibration we report the all-pair indicator at each count and its conjunction over the seven checkpoints. Exact two-sided Clopper–Pearson intervals describe the probability of run coverage for each fixed model; they do not estimate coverage conditional on the unobservable good event. Means and their uncertainty are computed across independent sampling seeds, not across correlated state pairs or nested checkpoints.

Mean widths, errors, propagation diagnostics, and ablation costs use 5000 seed-bootstrap resamples with analysis seed 899, and the 2.5th and 97.5th percentiles give the reported 95% intervals. Priority contrasts resample matched differences between seeds. The intervals are descriptive; we make no familywise significance claim across contrasts.

All allocated seeds remain in every summary. A run that does not complete within its step cap contributes the capped cost, so the endpoint is an acquisition cost under a restricted budget; every interaction run reported in this paper completed.

APPENDIX F NEURAL EXTENSION: TARGETS AND UPDATE ORDER

This appendix specifies how the finite-state quantities can be approximated with neural networks on rich observations. It fixes the targets, the parametrization, and the update order of Algorithm 2. The specification is not evaluated in Section VIII; ensemble disagreement, a shared encoder, and EMA targets do not by themselves give simultaneous model confidence or joint convergence.

A. FROZEN BEHAVIORAL TARGETS

Let $z = f_{\bar{\theta}}(o)$. A masked bootstrap ensemble with randomized priors [42] predicts bounded mean rewards $\hat{r}_k(z,a)$ and stochastic successor laws $\hat{P}_k(\cdot \mid z, a)$. The likelihood models aleatoric randomness, and differences between heads serve as a proxy for epistemic uncertainty. We use the bounded, frozen ground cost

$$h_{\bar{\theta}}(o,o') = R_{\max} \min\{1, \|f_{\bar{\theta}}(o) - f_{\bar{\theta}}(o')\|_2\}. \tag{77}$$

With the same action at both observations, set

$$\begin{aligned} q_k(o,o',a) &= (1-\gamma)|\hat{r}_k(z,a) - \hat{r}_k(z',a)| \\ &\quad + \gamma \,\mathrm{OT}_{h_{\bar{\theta}}}(\hat{P}_k, \hat{P}'_k), \end{aligned} \tag{78}$$

$$M(o,o') = \max_a K_{ens}^{-1} \sum_k q_k(o,o',a),$$

$$\hat{b}_a(o,o') = \beta_{cal}\, \mathrm{sd}_k(q_k(o,o',a)) + \gamma \nu_{OT}. \tag{79}$$

Bounded rewards and a bounded ground cost give $0 \le M \le R_{\max}$. The term ν_{OT} accounts for solver error, and β_{cal} is selected on a disjoint calibration split. Neither quantity substitutes for model coverage of unseen states. A calibration report must state the nominal level, the sampling unit, the sample size, and any distribution shift; marginal coverage on held-out data does not establish adaptive, simultaneous coverage.

For continuous actions, we fix a policy $\bar{\pi}$ within each epoch and use its induced reward and successor law. This defines a one-action policy metric $d^{\bar{\pi}}$, and the maximum over actions in the width critic below is dropped. A finite discretization of the actions instead defines a restricted-action metric. Neither construction equals the supremum over the continuous action set, and changing $\bar{\pi}$ changes the target metric.

B. A BOUNDED WIDTH CRITIC AND EXPLICIT ENDPOINTS

We use symmetric features $(|z - z'|, (z + z')/2, a)$ and an MLP g_ψ, with

$$Q^W_\psi(o, o', a) = R_{\max}\,\mathrm{sigmoid}(g_\psi), \qquad W_\psi = \max_a Q^W_\psi. \tag{80}$$

All self-pair outputs are set to zero. The parametrization enforces symmetry and boundedness, but neither the triangle inequality nor statistical validity. The surrogate endpoints, together with their midpoint and width, are

$$\begin{aligned} L_\psi &= \max\{0, M - W_\psi/2\}, \\ H_\psi &= \min\{R_{\max}, M + W_\psi/2\}, \\ \bar{d}_\psi &= (L_\psi + H_\psi)/2, \qquad w_\psi = H_\psi - L_\psi. \end{aligned} \tag{81}$$

Self-pair endpoints are zero. Note that W_ψ estimates a majorant, whereas w_ψ is the width of the clipped interval; the two differ. Before the first fit we use $M = R_{\max}/2$ and $W = R_{\max}$ off the diagonal, which gives $[L, H] = [0, R_{\max}]$.

On sampled successor supports, we optimize a feasible coupling $\hat{\Lambda}^a$ under the frozen lower endpoint L of the preceding epoch. With gradients stopped through every target quantity, the bounded target is

$$\begin{aligned} Y^W_a = \min\big\{R_{\max}, 2\hat{b}_a + \gamma\xi_\Lambda \\ + \gamma\mathbb{E}_{(u,v)\sim\hat{\Lambda}^a} W_{\bar{\psi}}(u, v)\big\}. \end{aligned} \tag{82}$$

The critic minimizes $\mathbb{E}[(Q^W_\psi - Y^W_a)^2]$. The maximum over actions and the coupling expectation mirror Proposition 7; independent successors or a deterministic mean successor would define different targets. If the feasible coupling is suboptimal by ξ_Λ under the lower cost, then

$$\mathrm{OT}_{d^+}(\hat{P}, \hat{P}') - \mathrm{OT}_{d^-}(\hat{P}, \hat{P}') \le \langle \hat{\Lambda}^a, \Delta\rangle + \xi_\Lambda, \tag{83}$$

which is why the source in (82) contains $\gamma\xi_\Lambda$. When ξ_Λ cannot be bounded, the target is an empirical approximation. A certified critic would further require a uniform Bellman residual and valid model sources; a small squared error averaged over replay does not suffice.

C. NORMALIZED REPRESENTATION LEARNING

The encoder is fitted to either certified or surrogate endpoints, and which one is used must be recorded. Put $\tilde{d} = \bar{d}/R_{\max}$, $\tilde{w} = w/R_{\max}$, and $D_\theta = \|f_\theta(o) - f_\theta(o')\|_2$. With $\zeta > 0$,

$$\begin{aligned} \mathcal{L}_{rep} = \tau_t\mathbb{E}_{temp}\frac{(D_\theta - \tilde{d})^2}{\zeta^2 + \tilde{w}^2} \\ + \mathbb{E}_{strat}\frac{(D_\theta - \tilde{d})^2}{\zeta^2 + \tilde{w}^2}. \end{aligned} \tag{84}$$

Both terms regress onto the same frozen target, as required by (40). Stratified replay balances the cells that have been identified; it cannot balance states that have not. For EMA averages m_t and w_t of the normalized midpoint and width, set

$$\tau_t = \begin{cases} 1, & m_t + w_t = 0, \\ \mathrm{clip}(m_t/(m_t + w_t), \tau_{\min}, 1), & m_t + w_t > 0. \end{cases} \tag{85}$$

The temperature weights the temporal loss and never rescales a pairwise target. For fixed positive m_t it decreases as uncertainty grows, so the global term dominates while the geometry is uncertain.

Proposition 24 (Reward-scale equivariance)**.** *Under $r \mapsto cr$ and $R_{\max} \mapsto cR_{\max}$ with $c > 0$, with reward radii scaled accordingly and transition radii unchanged, the exact endpoints, width, and source all scale by c. Their normalized versions, and hence* (84)*, are unchanged for the same normalized encoder.*

Proof. Transport is homogeneous in its ground cost, so $\mathcal{T}^{\pm}_{scaled}[ch] = c\mathcal{T}^{\pm}[h]$, and uniqueness gives the scaled fixed points. Dividing by $cR_{\max}$ proves the claim. □

Identical optimizer trajectories additionally require consistently normalized outputs, regularizers, rewards, and solver tolerances.

D. NAVIGATION, TRAINING ORDER, AND SCOPE

The agent keeps a memory C_t of observation anchors, refreshes their embeddings at each target epoch, and evaluates the novelty of a new observation before inserting it:

$$u_t(o) = \min_{c\in C_t}\{D_{\bar{\theta}}(o, c) + \beta_u W_\psi(o, c)/R_{\max}\}. \tag{86}$$

This quantity plays the role of the optimistic novelty $\min_c d^+(s, c)$ in the finite-state construction. The agent selects a frontier goal, navigates to it with a goal-conditioned policy trained with hindsight relabeling [43], and then explores locally; returning before exploring follows the control pattern of Go-Explore [44]. The task policy may be trained on

$$r^{train}_t = r^{ext}_t/R_{\max} + \beta_t W_\psi(o_t, c_t)/R_{\max}, \tag{87}$$

where c_t is the nearest previous anchor. Because the bonus changes over time, it is not a policy-invariant potential reward, and it must be disabled when extrinsic performance is evaluated. SAC and PPO [45], [46] are compatible choices

Algorithm 2 Neural EBG: specified extension, without an empirical certificate

Input: Encoder and targets, masked model ensemble, bounded pair critic, replay, anchors, calibration split, goal and task policies.

Set surrogate intervals to $[0, R_{\max}]$ off the diagonal.
for each target epoch **do**
  Refresh anchors; freeze the epoch policy and preceding lower proxy.
  Select a frontier by (86); navigate and explore.
  Fit bounded reward and stochastic successor heads with bootstrap masks.
  Freeze heads; form M, calibrated proxy sources, and feasible couplings.
  Update the width critic by (82).
  Freeze endpoints (81); fit (84).
  Update loss weight, policies, and EMA targets.
  Record calibration, failed navigation, model/solver error, and compute.
end for

of policy learner; the finite-state study reports no result for either.

The main failure mode is collapse: unseen states can be mapped onto familiar embeddings, and then all heads agree, wrongly. The specification does not solve latent-state identification. A certified decoder would make the quotient composition of Corollary 18 available; without one, that guarantee does not transfer to the extension.

E. TRANSPORT ERROR AND COMPUTATION

For finite supports, entropy-regularized transport is

$$\mathrm{OT}_h^{\lambda}(p,q) = \min_{\Lambda \in \Pi(p,q)} \{\langle \Lambda, h\rangle + \lambda\, \mathrm{KL}(\Lambda \| p \otimes q)\}. \quad (88)$$

Sinkhorn iterations [47], [48] approximate this objective, whose bias satisfies

$$0 \le \mathrm{OT}_h^{\lambda} - \mathrm{OT}_h \le \lambda \min\{H(p), H(q)\}. \quad (89)$$

The upper bound follows by evaluating an unregularized optimal coupling, whose mutual information is at most the entropy of either marginal. Feasible primal plans and unregularized dual potentials together give a direct numerical bracket. Debiased Sinkhorn divergences [49], [50] need not preserve the γ-contraction when the ground cost varies; a direct three-term bound has Lipschitz factor at most two.

Dense neural couplings cost $O(B_{pair} A_{eval} K_{ens} m^2 n_S)$ for m successor samples and n_S Sinkhorn iterations, and shared plans need their own error checks. Finite sparse plans can instead be stored in $O(AS^2\Gamma^2)$ space; with two successors per row, the finite-state implementation applies four coefficients per plan without forming a dense AS^4 transition tensor. Resource comparisons must include model fitting, calibration, anchor refresh, and navigation.

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